

Pharmacogenomics: predicted response to 632 drugs

Genetic response predicted to 632 drugs for frequent use, by therapeutic category
Report generated in October 2 of 2026

Genetic basis: Crude DNA (chip, GRCh38, 684 967 SNPs) + local imputation (Beagle 5.5, public reference panel of genomes complete to 30x; only genotypes with DR2 ≥ 0.8). Genetic data did not leave the local computer.

Evidence base: CPIC (official API, recommendations by phenotype), DPWG (KNMP), FDA tables of pharmacogenomic biomarkers and pharmacogenetic associations, FDA/EMA labels and levels of evidence ClinPGx/PharmGKB.

List of medicines: ClinCalc DrugStats Top 300 (2024; MEPS) unfolded by active substance + 338 additions (frequent use, underrepresented categories — oncology, anesthesia, anti-infectives, HIV/hepatitis, psychiatry, immunosuppressants, analgesics — and drugs with PGx guideline) + current medication.

Important: This is a pharmacogenetic reading of chip data, not a diagnosis or a prescription. All indications are suggestion (not applied) and should be decided with the doctor. Before a narrow margin drug, confirm with clinical pharmacogenetic test (especially CYP2D6 with CNV and HLA typing).

red avoid/alternative **orange** dose adjustment **yellow** monitoring/confirming **green** normal (relevant gene evaluated) **gray** indeterminate / not applicable **compact list** without known pharmacogenetic interaction ★ current medication · ☆ suspended

Contents	
1. Executive summary.....	2
2. Phenotype per gene	9
3. Medicines by therapeutic category	11
3.1 Cardiovascular	11
3.2 Anticoagulants and antiaggregants	14
3.3 Analgesics, NSAIDs and opioids	16
3.4 Psychiatry	19
3.5 Neurology.....	24
3.6 Anti-infective.....	27
3.7 Oncology	31
3.8 Anesthesia and perioperative	33
3.9 Endocrine and diabetes	35
3.10 Gastroenterology	37
3.11 Respiratory and allergy	39
3.12 Immunology, rheumatology and transplantation	41
3.13 Urology and sexual health	43
3.14 Dermatology	45
3.15 Hormones and corticosteroids	46
3.16 Other (vitamins, minerals, ophthalmology, hematology, dependencies).....	47
4. Limitations.....	48
5. Sources and methods.....	48

1. Executive summary

Class	Drug number	%
Avoid / alternative	2	0.3%
Adjust dose	16	2.5%
Monitoring / confirm	35	5.5%
Undetermined / not applicable	8	1.3%
Normal	226	35.8%
No known pharmacogenetic interaction	345	54.6%
Total analyzed	632	100%

Fast reading: 53 medicines request attention (avoid, adjust or monitor). Almost all are explained by four findings: CYP2C9 *1/*3 (NSAIDs, coumarins, phenytoin, siponimod, fluvastatin, sulfonylureas), VKORC1 –1639 T/T (coumarins), NAT2 slow acetylator (isoniazide, hydralazine, sulfonamides) and CYP3A5 expresser (tacrolímus, sirolimus). The opioids activated by CYP2D6 and tamoxifen are yellow only because the chip does not see duplications/deletions of CYP2D6. None of the drugs you currently take have a guideline that orders to avoid or change the dose. Two are yellow: sirolimus (CYP3A5 expresser, calibrate with dosage) and isotretinoin (hepatic genetic).

Avoid / choose alternative (2)

Medication	Category	Genes (state)	Predicted response	What to do (guideline summary)
piroxicam	Analyses	CYP2C9: IM (*1/*3)	avoid – choose alternative	CPIC 2020 (AS 1.0): choosing a NSAID not dependent on CYP2C9 or short half-life (very long half-lived → accumulation and risk GI/renal/CV).
tenoxicam	Analyses	CYP2C9: IM (*1/*3)	avoid – choose alternative	CPIC 2020 (AS 1.0): choosing a NSAID not dependent on CYP2C9 or short half-life (very long half-lived → accumulation and risk GI/renal/CV).

Adjust dose (16)

Medication	Category	Genes (state)	Predicted response	What to do (guideline summary)
celecoxib	Analyses	CYP2C9: IM (*1/*3)	higher exposure → lower dose	CPIC 2020 (AS 1.0): start with the lowest recommended dose and holder with caution; course as short as possible. FDA label: *3 carriers with increased exposure.
flurbiprofen	Analyses	CYP2C9: IM (*1/*3)	higher exposure → lower dose	CPIC 2020 (AS 1.0): start with the lowest recommended dose, titer with caution, short course. Avoid chronic use. (Also sold without revenue.)
ibuprofen	Analyses	CYP2C9: IM (*1/*3)	higher exposure → lower dose	CPIC 2020 (AS 1.0): start with the lowest recommended dose, titer with caution, short course. Avoid chronic use. (Also sold without revenue.)
lornoxicam	Analyses	CYP2C9: IM (*1/*3)	higher exposure → lower dose	CPIC 2020 (AS 1.0): start with the lowest recommended dose, titer with caution, short course. Avoid chronic use. (Also sold without revenue.)

Medication	Category	Genes (state)	Predicted response	What to do (guideline summary)
meloxicam	Analyses	CYP2C9 : IM (*1/*3)	higher exposure → 50% of the dose	CPIC 2020 (AS 1.0): start with 50% of the lowest recommended dose and do not exceed 50% of the maximum dose, or choose alternative. Long half.
isoniazid	Anti-infective	NAT2 : slow PNPLA3/Pi*MZ (liver) : unfavourable	higher exposure → hepatotoxicity	Slow acetylators have higher levels and more hepatotoxicity and neuropathy (FDA label; PharmGKB 1B). NAT2-guided schemas use a lower dose in slow ones. Add pyridoxin and tight hepatic surveillance (there are also unfavourable liver genetics).
acenocumarol	Anticoagulants	CYP2C9 : IM (*1/*3) VKORC1 : TT (sensitive)	adjust dose (very high sensitivity)	DPWG: VKORC1 –1639 AA (T/T) → start with ~50% of the standard dose and measure INR more frequently; CYP2C9*3 adds sensitivity especially in the induction phase. To prefer direct anticoagulant if possible (suggestion, not applied).
phenprocumon	Anticoagulants	CYP2C9 : IM (*1/*3) VKORC1 : TT (sensitive)	dose adjustment (high sensitivity)	DPWG: VKORC1 –1639 AA → ~50% of the initial dose and INR more frequent. CYP2C9*3 with minor effect on this coumarin.
warfarin	Anticoagulants	CYP2C9 : IM (*1/*3) VKORC1 : TT (sensitive) CYP4F2 : normal	adjust dose (very high sensitivity)	CPIC 2017: dose by genetic algorithm. With CYP2C9 *1/*3 + VKORC1 –1639 T/T, the FDA label table predicts a maintenance dose of ~0.5–2 mg/day (vs ~5 mg). Start low, INR tight in the first weeks. If the indication allows, prefer a direct anticoagulant (suggestion, not applied).
fluvastatin	Cardiovascular	CYP2C9 : IM (*1/*3) SLCO1B1 : increased function	higher exposure → ≤40 mg/day	CPIC 2022 CYP2C9 IM → initial dose ≤40 mg/day; if needed more, another statin or combination. SLCO1B1 of increased function does not change this.
hydralazine	Cardiovascular	NAT2 : slow	higher exposure → lower dose	CPIC: slow acetylator → initial total dose 40–75 mg/day, titer with care; caution with ≥200 mg/day (more induced lupus and adverse effects).
tacrolimus (systemic; the topic is not relevant PGx)	Immunology	CYP3A5 : expressor (*1/*3) CYP3A4 : normal	lower exposure → initial dose 1.5–2	CPIC 2015: CYP3A5 (*1/*3) → increase the initial dose 1.5–2× (without exceeding the maximum of the label) and guide by levels. Only relevant for systemic tacrolimus (transplantation, autoimmunity); the topic is not affected.
amifampridin	Neurology	NAT2 : slow	higher exposure → lower initial dose	FDA label: slow metabolizers NAT2 → start with the lowest recommended daily dose.
phenytoin	Neurology	CYP2C9 : IM (*1/*3) HLA-B*15:02 : n/a (very low priority)	higher exposure → ~25% less maintenance	CPIC2020: 1.the usual dose/load; maintenance ~25% below the usual, adjusted for serum levels. HLA-B*15:02 not evaluated, but unlikely in the reference population.
phosphenitoin	Neurology	CYP2C9 : IM (*1/*3) HLA-B*15:02 : n/a (very low priority)	higher exposure → ~25% less maintenance	CPIC2020: 1.the usual dose/load; maintenance ~25% below the usual, adjusted for serum levels. HLA-B*15:02 not evaluated, but unlikely in the reference population.
siponimod	Neurology	CYP2C9 : IM (*1/*3)	dose adjustment (maintenance 1 mg)	FDA/EMA labels: mandatory CYP2C9 genotyping. *1/*3 → maintenance dose 1 mg/day (instead of 2 mg); *3/*3 contraindicated.

Monitor / confirm before using (35)

Medication	Category	Genes (state)	Predicted response	What to do (guideline summary)
aceclofenac	Analyses	CYP2C9 : IM (*1/*3)	exposure possibly greater	CPIC 2020: no formal recommendation (insufficient evidence), but CYP2C9 is a relevant route. Use the lowest effective dose and the shortest course; avoid chronic use.
codeine	Analyses	CYP2D6 : Probable NM (without CNV) SLC22A1 (OCT1) : normal	probable normal - confirm CYP2D6 with CNV	CPIC 2021 NM → label dose. The main risk (death due to excess morphine) is ultrafast, and a duplication of CYP2D6 is not visible in the chip. Before regular use, confirm CYP2D6 with clinical test with CNV. Normal OCT1.

Medication	Category	Genes (state)	Predicted response	What to do (guideline summary)
diclofenac	Analyses	CYP2C9 : IM (*1/*3)	exposure possibly greater	CPIC 2020: no formal recommendation (insufficient evidence), but CYP2C9 is a relevant route. Use the lowest effective dose and the shortest course; avoid chronic use.
indomethacin	Analyses	CYP2C9 : IM (*1/*3)	exposure possibly greater	CPIC 2020: no formal recommendation (insufficient evidence), but CYP2C9 is a relevant route. Use the lowest effective dose and the shortest course; avoid chronic use.
methadone	Analyses	CYP2B6 : RM OPRM1 : A/G	possibly lower levels	Rapid CYP2B6 is associated with lower concentrations of methadone (PharmGKB 2A); CPIC without recommendation. Titular by response and ECG (QT).
naproxen	Analyses	CYP2C9 : IM (*1/*3)	exposure possibly greater	CPIC 2020: no formal recommendation (insufficient evidence), but CYP2C9 is a relevant route. Use the lowest effective dose and the shortest course; avoid chronic use.
nimesulide	Analyses	CYP2C9 : IM (*1/*3)	exposure possibly greater	CPIC 2020: no formal recommendation (insufficient evidence), but CYP2C9 is a relevant route. Use the lowest effective dose and the shortest course; avoid chronic use.
paracetamol	Analyses	UGT1A1 : IM (*1/*28) PNPLA3/Pi*MZ (liver) : unfavourable	monitoring (liver)	No PGx guideline. UGT1A1*28 (some minor glucuronidation) and unfavourable liver genetics (PNPLA3, MBOAT7, Pi*MZ probable): the genetic plan suggests ≤2 g/day (suggestion, not applied).
parecoxib	Analyses	CYP2C9 : IM (*1/*3)	exposure possibly greater	CPIC 2020: no formal recommendation (insufficient evidence), but CYP2C9 is a relevant route. Use the lowest effective dose and the shortest course; avoid chronic use.
tramadol	Analyses	CYP2D6 : Probable NM (without CNV) SLC22A1 (OCT1) : normal	probable normal - confirm CYP2D6 with CNV	CPIC 2021 NM → label dose. Ultrafast (duplication, not visible in the chip) are at risk of toxicity (respiratory depression, serotonergic syndrome). Normal OCT1 (normal exposure to active metabolite). Confirm CYP2D6 with CNV before regular use.
dapsone	Anti-infective	NAT2 : slow G6PD : normal	more hematological toxicity possible	Normal G6PD → with no reason to avoid (CPIC 2022). Slow acetylator: higher levels and more adverse effects (metemoglobinemia) described. HLA-B*13:01 not evaluated.
sulfamethoxazole (with trimethoprim)	Anti-infective	NAT2 : slow G6PD : normal	more hypersensitivity possible	Slow acetylators have more hypersensitivity reactions to sulfonamides (FDA label NAT); PharmGKB). Normal G6PD → without restriction of hemolysis (CPIC 2022).
procainamide	Cardiovascular	NAT2 : slow	higher exposure → plus induced lupus	FDA label: slow acetylators accumulate procainamide and have more induced lupus. Monitoring levels and ANA.
acitretin	Dermatology	PNPLA3/Pi*MZ (liver) : unfavourable	monitoring (liver)	No PGx guideline. Infavourable liver genetic context (PNPLA3, MBOAT7, probable Pi*MZ): is not a PGx guideline, but justifies surveillance of transaminases.
★ isotretinoin	Dermatology	CYP2C8 : normal CYP2C9 : IM (*1/*3) CYP3A4 : normal PNPLA3/Pi*MZ (liver) : unfavourable	compatible - monitoring liver/lipids	No PGx guideline. Main route (CYP2C8) normal; CYP2C9*3 can slightly increase exposure. Infavourable liver genetic context (PNPLA3, MBOAT7, probable Pi*MZ): is not a PGx guideline, but justifies surveillance of transaminases. Daily photoprotection (p88 nevus; isotretinoin).
glibenclamide	Endocrine	CYP2C9 : IM (*1/*3) G6PD : normal	higher exposure → more hypoglycemia	CYP2C9*3 increases sulfonylurea exposure (PharmGKB 2A-3); DPWG without mandatory action. Start low and monitor blood glucose. Normal G6PD: no risk of hemolysis.
gliclazide	Endocrine	CYP2C9 : IM (*1/*3) G6PD : normal	higher exposure → more hypoglycemia	CYP2C9*3 increases sulfonylurea exposure (PharmGKB 2A-3); DPWG without mandatory action. Start low and monitor blood glucose. Normal G6PD: no risk of hemolysis.

Medication	Category	Genes (state)	Predicted response	What to do (guideline summary)
glimepiride	Endocrine	CYP2C9 : IM (*1/*3) G6PD : normal	higher exposure → more hypoglycemia	CYP2C9*3 increases sulfonylurea exposure (PharmGKB 2A-3); DPWG without mandatory action. Start low and monitor blood glucose. Normal G6PD: no risk of hemolysis.
glipizide	Endocrine	CYP2C9 : IM (*1/*3) G6PD : normal	higher exposure → more hypoglycemia	CYP2C9*3 increases sulfonylurea exposure (PharmGKB 2A-3); DPWG without mandatory action. Start low and monitor blood glucose. Normal G6PD: no risk of hemolysis.
somatropin (growth hormone)	Endocrine	TERT/melanoma (context)	precaution (oncologic genetic context)	It is not pharmacogenetic interaction: the genetic plan suggests avoiding GH/IGF-1 by risk TERT alleles and nevi in the 88th percentile (melanoma by the chip is p6, 5/p25) (suggestion, not applied; weak evidence).
tolbutamide	Endocrine	CYP2C9 : IM (*1/*3) G6PD : normal	higher exposure → more hypoglycemia	CYP2C9*3 increases sulfonylurea exposure (PharmGKB 2A-3); DPWG without mandatory action. Start low and monitor blood glucose. Normal G6PD: no risk of hemolysis.
sulfasalazine	Gastroenterology	NAT2 : slow G6PD : normal	more adverse effects	Slow acetylators accumulate sulfapyridine: more nausea, headache, hemolysis and skin reactions (FDA label). Starting low and full-time. Normal G6PD.
leflunomide	Immunology	PNPLA3/Pi*MZ (liver) : unfavourable	monitoring (liver)	No PGx guideline. Infavourable liver genetic context (PNPLA3, MBOAT7, probable Pi*MZ): is not a PGx guideline, but justifies surveillance of transaminases.
methotrexate	Immunology	MTHFR : 677CT/1298AC SLCO1B1 : increased function	toxicity possibly something greater (weak evidence)	MTHFR 677 C/T: weak and inconsistent association with more toxicity (CPIC level C; DPWG without action). Support folate in the form 5-MTHF (non-folic acid; MTHFR 677CT/1298AC) and usual surveillance (hemogram, liver). Infavourable liver genetic context (PNPLA3, MBOAT7, probable Pi*MZ): is not a PGx guideline, but justifies surveillance of transaminases.
★ sirolimus (rapamycin)	Immunology	CYP3A5 :expressor (*1/*3) CYP3A4 : normal	exposure possibly lower per mg	No guideline for sirolimus. In transplantation, CYP3A5 expressers have higher clearance and lower levels per mg (moderate evidence). Normal CYP3A4. A sirolimus dosage to calibrate the dose (suggestion, not applied).
deutetrabenazine	Neurology	CYP2D6 : Probable NM (without CNV)	probable normal – label requires genotyping above certain doses	FDA label: genotipar CYP2D6 before high doses (pimozide >4 mg/day; tetrabenazine >50 mg/day). With probable NM the usual titration is applied, but confirm with clinical test with CNV.
tetrabenazine	Neurology	CYP2D6 : Probable NM (without CNV)	probable normal – label requires genotyping above certain doses	FDA label: genotipar CYP2D6 before high doses (pimozide >4 mg/day; tetrabenazine >50 mg/day). With probable NM the usual titration is applied, but confirm with clinical test with CNV.
belinostat	Oncology	UGT1A1 : IM (*1/*28)	exposure something bigger	FDA label: *28/*28 → reduced initial dose; *1/*28 without adjustment, monitoring toxicity.
irinotecan	Oncology	UGT1A1 : IM (*1/*28) DPYD : normal (AS 2)	something greater risk of neutropenia	FDA label: *1/*28 (IM) has increased risk of neutropenia; dose reduction recommended only for *28/*28. DPWG: *1/*28 without action. To monitor the blood count closely.
nilotinib	Oncology	UGT1A1 : IM (*1/*28) HLA-B*57:01 : negative	hyperbilirubinemia more likely	FDA labels: UGT1A1*28 increases hyperbilirubinemia (benign, but confuses liver surveillance). For pazopanib, HLA-B*57:01 negative (less risk of high ALT).
pazopanib	Oncology	UGT1A1 : IM (*1/*28) HLA-B*57:01 : negative	hyperbilirubinemia more likely	FDA labels: UGT1A1*28 increases hyperbilirubinemia (benign, but confuses liver surveillance). For pazopanib, HLA-B*57:01 negative (less risk of high ALT).
satituzumab govitecan	Oncology	UGT1A1 : IM (*1/*28)	something greater risk of neutropenia	FDA label: *28/*28 with more neutropenia; heterozygotes with intermediate risk. Monitoring blood count.

Medication	Category	Genes (state)	Predicted response	What to do (guideline summary)
tamoxifen	Oncology	CYP2D6 : Probable NM (without CNV) F5/F2 : non carrier	probable normal - confirm CYP2D6 with CNV	CPIC 2018 NM → tamoxifen 20 mg/day and avoid moderate/strong inhibitors of CYP2D6 (paroxetine, fluoxetine, bupropion). If an *5 deletion turned it into IM, the recommendation would change (major dose or aromatase inhibitor) → confirm with CNV before. F5/F2 normal.
eliglustat	Other (vitamin	CYP2D6 : Probable NM (without CNV)	Mandatory clinical genotyping	FDA/EMA label: the CYP2D6 phenotype defines the dose (NM: 84 mg 2×/day). The chip does not exclude CNV → needs clinical testing.
pimozide	Psychiatry	CYP2D6 : Probable NM (without CNV)	probable normal - label requires genotyping above certain doses	FDA label: genotipar CYP2D6 before high doses (pimozide >4 mg/day; tetrabenazine >50 mg/day). With probable NM the usual titration is applied, but confirm with clinical test with CNV.

Undetermined / not applicable (8)

Medication	Category	Genes (state)	Predicted response	What to do (guideline summary)
amikacin	Anti-infective	MT-RNR1 : indeterminate	indeterminate (mitochondrial ototoxicity)	CPIC 2021: carriers of m.1555A>G/m.1494C>T should avoid aminoglycosides without any alternative. The MT-RNR1 state is not determinable in this chip. Without result, CPIC indicates normal use, short course and auditory surveillance; if repeated or prolonged use is predicted, clinical testing of m.1555A>G (suggestion, not applied). Oral/topic neomycin has low absorption.
streptomycin	Anti-infective	MT-RNR1 : indeterminate	indeterminate (mitochondrial ototoxicity)	CPIC 2021: carriers of m.1555A>G/m.1494C>T should avoid aminoglycosides without any alternative. The MT-RNR1 state is not determinable in this chip. Without result, CPIC indicates normal use, short course and auditory surveillance; if repeated or prolonged use is predicted, clinical testing of m.1555A>G (suggestion, not applied). Oral/topic neomycin has low absorption.
gentamicin	Anti-infective	MT-RNR1 : indeterminate	indeterminate (mitochondrial ototoxicity)	CPIC 2021: carriers of m.1555A>G/m.1494C>T should avoid aminoglycosides without any alternative. The MT-RNR1 state is not determinable in this chip. Without result, CPIC indicates normal use, short course and auditory surveillance; if repeated or prolonged use is predicted, clinical testing of m.1555A>G (suggestion, not applied). Oral/topic neomycin has low absorption.
neomycin	Anti-infective	MT-RNR1 : indeterminate	indeterminate (mitochondrial ototoxicity)	CPIC 2021: carriers of m.1555A>G/m.1494C>T should avoid aminoglycosides without any alternative. The MT-RNR1 state is not determinable in this chip. Without result, CPIC indicates normal use, short course and auditory surveillance; if repeated or prolonged use is predicted, clinical testing of m.1555A>G (suggestion, not applied). Oral/topic neomycin has low absorption.
terbinafine	Anti-infective	HLA-A*33:01 : indeterminate PNPLA3/Pi*MZ (liver) : unfavourable	indeterminate (liver risk)	HLA-A*33:01 is associated with liver injury by oral terbinafine (PharmGKB) and is not evaluated here. Infavourable liver genetic context (PNPLA3, MBOAT7, probable Pi*MZ): is not a PGx guideline, but justifies surveillance of transaminases. To prefer topical treatment when possible (suggestion, not applied).
tobramycin	Anti-infective	MT-RNR1 : indeterminate	indeterminate (mitochondrial ototoxicity)	CPIC 2021: carriers of m.1555A>G/m.1494C>T should avoid aminoglycosides without any alternative. The MT-RNR1 state is not determinable in this chip. Without result, CPIC indicates normal use, short course and auditory surveillance; if repeated or prolonged use is predicted, clinical testing of m.1555A>G (suggestion, not applied). Oral/topic neomycin has low absorption.
lapatinib	Oncology	HLA-DRB1*07:01 : indeterminate	indeterminate	FDA label: HLA-DRB1*07:01/DQA1*02:01 are associated with hepatotoxicity; not evaluated in this chip.
ivacaftor (and combinations with elxacaftor/tezacaftor)	Respiratory	CFTR : not applicable	not applicable	Indicated only in cystic fibrosis with specific CFTR variants (CPIC 2014). It does not apply.

Pharmacogenetic card (for the clinical process)

Situation	Genotype	Indication (suggestion, not applied)
Anticoagulation	CYP2C9 *1/*3 + VKORC1 -1639 T/T + CYP4F2 C/C	To prefer direct anticoagulant (apixaban, rivaroxaban, edoxaban, dabigatran). If coumarin: a very low initial dose by genetic algorithm and tight INR.
Pain / anti-inflammatory	CYP2C9 *1/*3 (AS 1.0)	Piroxicam and tenoxicam: avoid. Meloxicam: 50% of the dose. Ibuprofen, celecoxib, flurbiprofen, lornoxicam: lower dose and short course. Avoid chronic NSAID.
Antiepileptics	CYP2C9 IM; HLA-A*31:01 neg. HLA-B*15:02 n/a	Phenytoin/phosphenytoin: ~25% less maintenance, by levels. Carbamazepine: standard dose (confirm A*31:01 if long therapy).
Tuberculosis / hydralazine / sulfonamides	NAT2 *5/*5 (slow acetylator)	Isoniazid: dose guided by NAT2 + hepatic surveillance. Hydralazine: 40-75 mg/day. Sulfasalazine, sulfamethoxazole, dapsone, procainamide: monitoring adverse effects.
Immunosuppression	CYP3A5 *1/*3 (expressor)	Tacrolimus: initial dose 1.5-2x with levels. Sirolimus: calibrate with dosage.
Opioides / tamoxifen / eligostat	CYP2D6 *1/*2 probable, CNV not evaluated	Codeine, tramadol, tamoxifen: standard dose if a clinical test with CNV confirms NM.
Oncology	Normal DPYD (13 var.); TPMT/NUDT15 normal; UGT1A1 *1/*28	Fluoropyrimidines and thiopurines: standard dose (repeat clinical DPD, as EMA recommends). Irinotecan/sacituzumab: monitoring neutropenia.
Statins	SLCO1B1 *1/*14 (augmented function); Normal ABCG2	All at the desired dose, except fluvastatin ≤40 mg/day (CYP2C9).
Anesthesia	RYR1/CACNA1S without variants; BCHE normal	Standard precautions for volatiles and suxamethonium (reduced risk, not absent).
Hemolysis / oxidants	Normal G6PD (hemizigote)	Primaquine, taphenoquine, rasburicase, dapsone, methylene blue: no genetic restriction.
Drop	HLA-B*58:01 negatable (imputed)	Allopurinol at the standard dose; clinical typing before starting is still good practice.
Aminoglycosides	MT-RNR1 not determinable	Short course and auditory surveillance; m.1555A>G test if repeated use.

Current medicine × genetics

Drug	Genes (state)	Interpretation	Comment
★ isotretinoin	CYP2C8: normal CYP2C9: IM (*1/*3) CYP3A4: normal PNPLA3/Pi*MZ (liver): unfavourable	compatible - monitoring liver/lipids	No PGx guideline. Main route (CYP2C8) normal; CYP2C9*3 can slightly increase exposure. Infavourable liver genetic context (PNPLA3, MBOAT7, probable Pi*MZ): is not a PGx guideline, but justifies surveillance of transaminases. Daily photoprotection (p88 nevus; isotretinoin).
★ sirolimus (rapamycin)	CYP3A5:expressor (*1/*3) CYP3A4: normal	exposure possibly lower per mg	No guideline for sirolimus. In transplantation, CYP3A5 expressers have higher clearance and lower levels per mg (moderate evidence). Normal CYP3A4. A sirolimus dosage to calibrate the dose (suggestion, not applied).
★ acarbose	—	normal	Without known PGx (acts in the intestine, it is almost not absorbed).
★ dapagliflozin	UGT1A9: normal	normal	Main normal UGT1A9 (without *3) → normal exposure.
★ ezetimiba	UGT1A1: IM (*1/*28)	normal	Glucuronidated (UGT1A1/1A3); *1/*28 without known clinical impact.
★ imeglumine	SLC22A2 (OCT2): A270S het. SLC22A1 (OCT1): normal	normal	Excreted by the kidney via OCT2/MATE; OCT2 A270S heterozygous, effect not studied and probably irrelevant. Normal OCT1.
☆ inclisirane	PCSK9 (informative)	normal	No actionable PGx (antibody/siRNA anti-PCSK9). PCSK9 without R46L, so without "natural protection" that reduces utility.
★ naltrexone (incl. low dose)	OPRM1: A/G	normal	OPRM1 118 A/G: some studies (alcoholism) give better response to carriers G; inconsistent evidence. CPIC without recommendation.

Drug	Genes (state)	Interpretation	Comment
★ oxytocin	OXTR (informative)	normal	No actionable PGx. OXTR rs53576 G/G is associated, inconsistently, with greater social sensitivity to oxytocin.
★ pitavastatin	SLCO1B1: increased function	normal (risk of non-augmented myopathy)	CPIC 2022 SLCO1B1 of increased function → desired dose, adjusted by disease guidelines.
★ retatrutide	GLP1R (informative)	normal	Peptides without CYP metabolism; without actionable PGx. (GLP1R variants associated with response only in exploratory studies.)
★ tadalafil	CYP3A4: normal	normal	Substrates of CYP3A4, which is normal (without *22). No actionable PGx.
★ telmisartan	UGT1A1: IM (*1/*28)	normal	Glucuronidated (UGT1A3); without actionable PGx.
★ bempedoic acid	ABCG2: normal HLA-B*58:01: probable negative	normal	No PGx guideline. Normal ABCG2 (without strong genetic predisposition to high urate). If ever it is necessary to allopurinol, see HLA-B*58:01.
Stack peptides (epitalon, pinealon, glutathione, SS-31, 5-amino-1MQ, MOTS-c, humanin, thymosine α1, thymalin, FOXO4-DRI)	—	without known PGx	No known metabolism genes. Non-pharmacological genetic notes (e.g. TERT/melanoma for epitalon) are in the genetic plane.

What this work adds to what was already known

- **OCT1 (SLC22A1) of normal function:** the four common variants of function loss (R61C, G401S, M420del, G465R) are absent (chip + imputation DR2 1.00). Normal exposure to tramadol, morphine, ondansetron, sumatriptan and metformin. *New.*
- **POLG:** A467T and W748S absent → without common genetic contraindications for valproate. *New.*
- **IFNL4** rs368234815 TT/TT (Imputed DR2 1.00), consistent with IFNL3 C/C (Favourable Interferon Response). *New.*
- **CYP2D6*9 absent** (Imputed, DR2 0.92), which reinforces the probable *1/*2. *17 is not attributable (DR2 0), but it is rare in the reference population. CNV remains to be evaluated. *New.*
- **CYP2A6** *2 and *9 absent (relevant to nicotine, letrozole and tegafur); the deletions remain invisible. *New.*
- **MT-RNR1:** the probe "M:1555-A-G" exists in the chip, but is mapped in rCRS 1553; the mitochondrial probes of this chip use ambiguous coordinates (the "M:3243-A-G" is also displaced). Conclusion: it remains undeterminable. *Clarification.*
- Now applied to each drug: F5 Leiden and prothrombin negative (estrogens, tamoxifen, thrombopoietic); Normal BCHE (suxamethonium, mivacurium); HLA-DRB1*15:01 negative (amoxicillin-clavulanate). *Already known.*
- Confirmed without change: CYP2C9 *1/*3, VKORC1 T/T, NAT2 slow, UGT1A1 *1/*28, CYP3A5 expresser, SLCO1B1 *1/*14, CYP2C19 *1/*1, CYP2B6 *2/*4, DPYD/TPMT/NUDT15/G6PD normal, HLA-B*57:01 negative, B*58:01 and A*31:01 probable negative.

2. Phenotype per gene

Gene	Genotype	Phenotype	Trust / origin
CYP2C9	*1/*3 (rs1057910 A/C)	intermediate metabolizer (AS 1.0)	high - direct chip
VKORC1	−1639 T/T (rs9923231); rs9934438 A/A	high sensitivity to coumarins	high - DR2 0.99 + chip imputed
CYP4F2	*1/*1 (rs2108622 C/C)	normal	high - direct chip
CYP2C19	*1/*1 (without *2, *3, *4, *17)	normal metabolizer	high - direct chip
CYP2D6	Probable *1/*2 (rs16947 A/G, rs1080985 C/G; without *3/*4/*6/*9/*10/*41)	Probable normal metabolizer (AS 2)	moderate - chip + imputation; CNV (deletion *5, duplications)
CYP3A5	*1/*3 (rs776746 C/T)	expresser (intermediate metabolizer)	high - direct chip
CYP3A4	*1/*1 (without *22, rs35599367 G/G)	normal	high - direct chip
CYP2B6	Probable *2/*4 (rs2279343 A/G, rs8192709 C/T; without *6/*18)	fast metabolizer	Moderate - direct chip; undetermined phase
CYP1A2	*1F heterozygous (rs762551 C/A)	intermediate inductibility	high - direct chip (weak clinical effect)
CYP2C8	without *3 (rs11572080 C/C) or *4	normal	high - chip + imputation DR2 1.00
CYP2A6	without *2 or *9; deletions (*4) not visible	probable normal	low - deletions not evaluated
SLCO1B1	*1/*14 (521 T/T; 388G and 463A heterozygotes)	increased function	high - direct chip
ABCG2	Normal Q141 (rs2231142 G/G)	normal function	high - direct chip
UGT1A1	*1/*28 ((TA)7 heterozygous; rs887829 C/T)	intermediate metabolizer (Gilbert heterozygote)	high - DR2 1.00 + chip imputed
UGT1A9	without *3; −2152 C/C	normal	high - direct chip
DPYD	without *2A, *13, c.2846A>T, HapB3 or Y186C (13 variants)	normal (AS 2) for the tested variants	high for the tested variants - chip
TPMT	*1/*1 (without *2, *3A, *3B, *3C)	normal metabolizer	high - direct chip
NUDT15	*1/*1 (without *3)	normal metabolizer	high - direct chip
G6PD	B (normal), hemizigote; without c.202G>A (rs1050828) or c.563C>T (rs5030868)	normal	high for the tested variants - chip
NAT2	Probable *5/*5 (341C, 481T, 803G homozygotes)	slow acetylator	high - direct chip (7 SNPs)
HLA-B*57:01	negative (rs2395029 T/T; imputed tag r2 1.00)	non carrier	high - tag chip + imputation
HLA-B*58:01	Probable negative (Imputational tag chr6:31360023, r2 1.00 in 378 genomes of the reference population, only 6 carriers)	not probable carrier	moderate - imputed tag
HLA-A*31:01	probable negative (rs1061235 A/A; imputed tag r2 1.00)	not probable carrier	moderate - tag chip + imputation
HLA-B*15:02	non-evaluable (non-informative tags in the reference population)	very low a priori probability in the reference population (<0.1%)	not determinable
HLA-A*33:01	non-evaluable (best r2 0.75 tag; DR2 0.66)	indeterminate	not determinable
HLA-DRB1*15:01	negative (impacted tags)	non carrier	moderate - imputed tags
HLA-DRB1*07:01	not evaluated	indeterminate	not determinable

Gene	Genotype	Phenotype	Trust / origin
MT-RNR1	not determinable: the chip 'M:1555-A-G' probe is mapped in rCRS 1553 (ambiguous mitochondrial coordinates); m.1494C>T not tested	indeterminate	not determinable (mtDNA is also not attributable)
RYR1/CACNA1S	34 + 2 pathogenic variants tested, none present	without identified susceptibility	partial - chip (only known variants)
BCHE	without atypical variant (rs1799807 T/T) or K (rs1803274 C/C)	normal	high - direct chip
IFNL3/IFNL4	rs12979860 C/C; rs8099917 T/T; IFNL4 rs368234815 TT/TT	favourable genotype	high - chip + imputation DR2 1.00
F5/F2	F5 Leiden (rs6025) C/C; prothrombin G20210A (rs1799963) G/G	normal (non-carrier)	high - direct chip
POLG	A467T (rs113994095) absent; W748S (rs113994097) absent	normal for common variants	high for these variants - chip + imputed DR2 1.00
CFTR	G551D absent; Untrustable F508del in imputation (DR2 0.68)	not applicable (without diagnosis of cystic fibrosis)	partial
SLC22A1 (OCT1)	without R61C, G401S, M420del or G465R	normal function	high - chip + imputation DR2 1.00
SLC22A2 (OCT2)	A270S heterozygous (rs316019 A/C)	tubular secretion possibly slightly reduced	high - chip (uncertain clinical effect)
MTHFR	677 C/T + 1298 A/C (probable compound heterozygous)	slightly reduced activity	high - direct chip (weak effect)
OPRM1	118 A/G (rs1799971)	G allele carrier	high - chip (inconsistent clinical effect)
COMT	Val158Met G/G (Val/Val; haplotype LPS/LPS)	high activity	high - chip (weak clinical effect)
CYB5R	rare variants not tested	indeterminate	not determinable
UGT2B17	deletion not visible	indeterminate	not determinable
PNPLA3/Pi*MZ (liver)	PNPLA3 and MBOAT7 risk; SERPINA1 Pi*MZ probable	hepatic genetic vulnerability	Moderate - chip + imputation (Pi*Z per probe + haplotype)

Colors of this table: orange : phenotype with dose consequences; yellow = attention or partial confidence; gray = non-determinable; green = normal. The genes OPRM1, COMT, MTHFR, CYP1A2 and OCT2 are informative (no actionable guideline).

3. Medicines by therapeutic category

In each category, the table shows drugs with relevant pharmacogenetic gene, ordered by severity. Below comes the compact list of those who do not have known pharmacogenetic interaction, with the main pathway of metabolism and, if it is an evaluated gene, its state in Sérgio F.

3.1 Cardiovascular — 70 medicines

dose adjustment: 2 · Monitoring: 1 · normal: 27 · without known pharmacogenetic interaction: 40

Medication	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
fluvastatin Adjust dose	CYP2C9 : IM (*1/*3) SLCO1B1 : increased function	higher exposure → ≤40 mg/day	CPIC 2022 CYP2C9 IM → initial dose ≤40 mg/day; if needed more, another statin or combination. SLCO1B1 of increased function does not change this.	CPIC 2022 (statins) <i>moderate (CPIC A)</i>	CYP2C9: high SLCO1B1: high
hydralazine Adjust dose	NAT2 : slow	higher exposure → lower dose	CPIC: slow acetylator → initial total dose 40–75 mg/day, titer with care; caution with ≥200 mg/day (more induced lupus and adverse effects).	CPIC (hydralazine–NAT2); label FDA <i>moderate (CPIC A)</i>	NAT2: high
procainamide Monitoring	NAT2 : slow	higher exposure → plus induced lupus	FDA label: slow acetylators accumulate procainamide and have more induced lupus. Monitoring levels and ANA.	FDA label <i>moderate</i>	NAT2: high
alirocumab Normal	PCSK9 (informative)	normal	No actionable PGx (antibody/siRNA anti-PCSK9). PCSK9 without R46L, so without "natural protection" that reduces utility.	Genetic plan <i>weak</i>	—
atorvastatin Normal	SLCO1B1 : increased function	normal (risk of non-augmented myopathy)	CPIC 2022 SLCO1B1 of increased function → desired dose, adjusted by disease guidelines.	CPIC 2022; DPWG; labels FDA <i>strong (CPIC A; PharmGKB 1A)</i>	SLCO1B1: high
candesartan Normal	CYP2C9 : IM (*1/*3)	normal (minor effect)	CYP2C9*3 slightly reduces the conversion of losartan into the active metabolite and increases slightly.	PharmGKB (3 level) <i>weak</i>	CYP2C9: high
carvedilol Normal	CYP2D6 : Probable NM (without CNV)	normal	CPIC 2024 (betablockers): metoprolol in NM → standard dose; carvedilol/nebivolol/propranolol without action for NM. FDA labels refer only to slow metabolizers. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	CPIC 2024; DPWG; labels FDA <i>moderate to strong (PharmGKB 1A for metoprolol)</i>	CYP2D6: moderate
isosorbide dinitrate/ mononitrate Normal	CYB5R : indeterminate	normal	FDA label: risk of CYB5R deficit methemoglobinemia (rare, not testable in the chip). No other PGx.	FDA label <i>weak</i>	CYB5R: non-determinable
evolocumab Normal	PCSK9 (informative)	normal	No actionable PGx (antibody/siRNA anti-PCSK9). PCSK9 without R46L, so without "natural protection" that reduces utility.	Genetic plan <i>weak</i>	—
★ ezetimiba Normal	UGT1A1 : IM (*1/*28)	normal	Glucuronidated (UGT1A1/1A3); *1/*28 without known clinical impact.	PharmGKB <i>weak</i>	UGT1A1: high
flecainide Normal	CYP2D6 : Probable NM (without CNV)	normal	DPWG: adjustments only for slow/intermediate/ultrarate metabolizers. Probable NM → usual dose with ECG. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	DPWG; label FDA (propafenone) <i>moderate (PharmGKB 1A)</i>	CYP2D6: moderate

Medication	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
☆ inclisirane Normal	PCSK9 (informative)	normal	No actionable PGx (antibody/siRNA anti-PCSK9). PCSK9 without R46L, so without "natural protection" that reduces utility.	Genetic plan <i>weak</i>	—
Come on. Normal	CYP2C9 : IM (*1/*3)	normal (minor effect)	CYP2C9*3 slightly reduces the conversion of losartan into the active metabolite and increases slightly.	PharmGKB (3 level) <i>weak</i>	CYP2C9: high
losartan Normal	CYP2C9 : IM (*1/*3)	normal (minor effect)	CYP2C9*3 slightly reduces the conversion of losartan into the active metabolite and increases slightly.	PharmGKB (3 level) <i>weak</i>	CYP2C9: high
lovastatin Normal	SLCO1B1 : increased function	normal (risk of non-augmented myopathy)	CPIC 2022 SLCO1B1 of increased function → desired dose, adjusted by disease guidelines.	CPIC 2022; DPWG; labels FDA <i>strong (CPIC A; PharmGKB 1A)</i>	SLCO1B1: high
mivacamten Normal	CYP2C19 : NM	normal	The CYP2C19 is a relevant route and is included in the label/guidelines, but the recommendations apply to intermediate/slow metabolizers. Normal metabolizer → usual dose.	Labels FDA/EMA; CPIC (level B/C) <i>moderate</i>	CYP2C19: high
metoprolol Normal	CYP2D6 : Probable NM (without CNV)	normal	CPIC 2024 (betablockers): metoprolol in NM → standard dose; carvedilol/nebivolol/propranolol without action for NM. FDA labels refer only to slow metabolizers. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	CPIC 2024; DPWG; labels FDA <i>moderate to strong (PharmGKB 1A for metoprolol)</i>	CYP2D6: moderate
mexylethin Normal	CYP2D6 : Probable NM (without CNV)	normal	DPWG: adjustments only for slow/intermediate/ultrarate metabolizers. Probable NM → usual dose with ECG. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	DPWG; label FDA (propafenone) <i>moderate (PharmGKB 1A)</i>	CYP2D6: moderate
nebivolol Normal	CYP2D6 : Probable NM (without CNV)	normal	CPIC 2024 (betablockers): metoprolol in NM → standard dose; carvedilol/nebivolol/propranolol without action for NM. FDA labels refer only to slow metabolizers. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	CPIC 2024; DPWG; labels FDA <i>moderate to strong (PharmGKB 1A for metoprolol)</i>	CYP2D6: moderate
nitroglycerin Normal	CYB5R : indeterminate	normal	FDA label: risk of CYB5R deficit methemoglobinemia (rare, not testable in the chip). No other PGx.	FDA label <i>weak</i>	CYB5R: non-determinable
★ pitavastatin Normal	SLCO1B1 : increased function	normal (risk of non-augmented myopathy)	CPIC 2022 SLCO1B1 of increased function → desired dose, adjusted by disease guidelines.	CPIC 2022; DPWG; labels FDA <i>strong (CPIC A; PharmGKB 1A)</i>	SLCO1B1: high
pravastatin Normal	SLCO1B1 : increased function	normal (risk of non-augmented myopathy)	CPIC 2022 SLCO1B1 of increased function → desired dose, adjusted by disease guidelines.	CPIC 2022; DPWG; labels FDA <i>strong (CPIC A; PharmGKB 1A)</i>	SLCO1B1: high
propafenone Normal	CYP2D6 : Probable NM (without CNV)	normal	DPWG: adjustments only for slow/intermediate/ultrarate metabolizers. Probable NM → usual dose with ECG. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	DPWG; label FDA (propafenone) <i>moderate (PharmGKB 1A)</i>	CYP2D6: moderate

Medication	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
propranolol Normal	CYP2D6 : Probable NM (without CNV)	normal	CPIC 2024 (betablockers): metoprolol in NM → standard dose; carvedilol/nebivolol/propranolol without action for NM. FDA labels refer only to slow metabolizers. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	CPIC 2024; DPWG; labels FDA <i>moderate to strong (PharmGKB 1A for metoprolol)</i>	CYP2D6: moderate
quinidine Normal	CYP2D6 : Probable NM (without CNV)	normal	The CYP2D6 is on the label/guidelines, but the recommendations apply to slow or ultrafast metabolizers. Probable NM → usual dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	FDA/EMA labels; CPIC 2021 (hydrocodone/oxycodone: no action for NM) <i>moderate</i>	CYP2D6: moderate
rosuvastatin Normal	SLCO1B1 : increased function ABCG2 : normal	normal	CPIC 2022 SLCO1B1 increased function + normal ABCG2 → desired dose.	CPIC 2022; DPWG <i>strong (CPIC A; PharmGKB 1A)</i>	SLCO1B1: high ABCG2: high
simvastatin Normal	SLCO1B1 : increased function	normal (risk of non-augmented myopathy)	CPIC 2022 SLCO1B1 of increased function → desired dose, adjusted by disease guidelines.	CPIC 2022; DPWG; labels FDA <i>strong (CPIC A; PharmGKB 1A)</i>	SLCO1B1: high
★ telmisartan Normal	UGT1A1 : IM (*1/*28)	normal	Glucuronidated (UGT1A3); without actionable PGx.	PharmGKB <i>weak</i>	UGT1A1: high
torasemide Normal	CYP2C9 : IM (*1/*3)	normal	CYP2C9 contributes to metabolism, but label recommendations apply only to slow metabolizers (*3/*3) or associations with inhibitors. Usual dose.	FDA/EMA labels <i>moderate</i>	CYP2C9: high
★ bempedoic acid Normal	ABCG2 : normal HLA-B*58:01 : probable negative	normal	No PGx guideline. Normal ABCG2 (without strong genetic predisposition to high urate). If ever it is necessary to allopurinol, see HLA-B*58:01.	Genetic plan <i>weak</i>	ABCG2: high HLA-B*58:01: moderate

No known pharmacogenetic interaction (40):

amloride (renal)	enalapril (sterases; renal)	nimodipine (CYP3A4) — CYP3A4: normal
amiodarone (CYP3A4; CYP2C8) — CYP3A4: normal; CYP2C8: normal	eplerenone (CYP3A4) — CYP3A4: normal	olmesartan (sterases; biliary)
amlodipine (CYP3A4) — CYP3A4: normal	spironolactone (deacetylation)	perindopril (sterases; renal)
atenolol (renal)	phenofibrate (sterases; UGT)	termosine (hepatic)
benazepril (sterases; renal)	finerenone (CYP3A4) — CYP3A4: normal	ramipril (sterases; renal)
bisoprolol (CYP3A4; renal) CYP3A4: normal	furosemide (renal; UGT1A9) UGT1A9: normal	ranolazine (CYP3A4; CYP2D6) — CYP3A4: normal; CYP2D6: Probable NM (without CNV)
bometanide (renal)	hydrochlorothiazide (renal)	sacubitril (with valsartan) (sterases)
captopril (renal)	indapamide (CYP3A4) — CYP3A4: normal	sotalol (renal)
clonidine (CYP2D6 (partial); renal) CYP2D6: Probable NM (without CNV)	ivabradine (CYP3A4) — CYP3A4: normal	triantherene (CYP1A2) — CYP1A2: *1F het.
chlorthalidone (renal)	labetalol (UGT)	trimetazidine (renal)
digoxin (renal; P-gp)	lercanidipine (CYP3A4) — CYP3A4: normal	valsartan (biliary)
diltiazem (CYP3A4) — CYP3A4: normal	lysinoiril (renal (no metabolism)	verapamil (CYP3A4) — CYP3A4: normal
dronedarone (CYP3A4) — CYP3A4: normal	minoxidil (SULT1A1)	ethyl esters of omega-3 acids (β-oxidation)
	nifedipine (CYP3A4) — CYP3A4: normal	

3.2 Anticoagulants and antiaggregants — 18 medicines

dose adjustment: 3 · normal: 11 · without known pharmacogenetic interaction: 4

Medication	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
acenocoumarol Adjust dose	CYP2C9 : IM (*1/*3) VKORC1 : TT (sensitive)	adjust dose (very high sensitivity)	DPWG: VKORC1 –1639 AA (T/T) → start with ~50% of the standard dose and measure INR more frequently; CYP2C9*3 adds sensitivity especially in the induction phase. To prefer direct anticoagulant if possible (suggestion, not applied).	DPWG (acenocoumarol–VKORC1/CYP2C9); CPIC (level B) <i>strong (DPWG; PharmGKB 1B)</i>	CYP2C9: high VKORC1: high
phenprocoumon Adjust dose	CYP2C9 : IM (*1/*3) VKORC1 : TT (sensitive)	dose adjustment (high sensitivity)	DPWG: VKORC1 –1639 AA → ~50% of the initial dose and INR more frequent. CYP2C9*3 with minor effect on this coumarin.	DPWG <i>strong (DPWG)</i>	CYP2C9: high VKORC1: high
warfarin Adjust dose	CYP2C9 : IM (*1/*3) VKORC1 : TT (sensitive) CYP4F2 : normal	adjust dose (very high sensitivity)	CPIC 2017: dose by genetic algorithm. With CYP2C9 *1/*3 + VKORC1 –1639 T/T, the FDA label table predicts a maintenance dose of ~0.5–2 mg/day (vs ~5 mg). Start low, INR tight in the first weeks. If the indication allows, prefer a direct anticoagulant (suggestion, not applied).	CPIC 2017; DPWG; label FDA (dose table) <i>strong (CPIC A; PharmGKB 1A)</i>	CYP2C9: high VKORC1: high CYP4F2: high
apixaban Normal	F5/F2 : non carrier CYP3A4 : normal	normal	Direct anticoagulants: without actionable PGx (the rivaroxaban label refers to F5 Leiden only in the context of thrombophilia; absent). They do not depend on CYP2C9/VKORC1, so they are the preferred option in the PGx card.	FDA labels; genetic plan <i>moderate</i>	F5/F2: high CYP3A4: high
cilostazol Normal	CYP2C19 : NM	normal	The CYP2C19 is a relevant route and is included in the label/guidelines, but the recommendations apply to intermediate/slow metabolizers. Normal metabolizer → usual dose.	Labels FDA/EMA; CPIC (level B/C) <i>moderate</i>	CYP2C19: high
clopidogrel Normal	CYP2C19 : NM	normal activation	CPIC 2022: normal metabolizer → clopidogrel 75 mg/day at the standard dose in all indications.	CPIC 2022; DPWG; FDA label (box alert for PM) <i>strong (CPIC A; PharmGKB 1A)</i>	CYP2C19: high
dabigatran Normal	F5/F2 : non carrier CYP3A4 : normal	normal	Direct anticoagulants: without actionable PGx (the rivaroxaban label refers to F5 Leiden only in the context of thrombophilia; absent). They do not depend on CYP2C9/VKORC1, so they are the preferred option in the PGx card.	FDA labels; genetic plan <i>moderate</i>	F5/F2: high CYP3A4: high
edoxaban Normal	F5/F2 : non carrier CYP3A4 : normal	normal	Direct anticoagulants: without actionable PGx (the rivaroxaban label refers to F5 Leiden only in the context of thrombophilia; absent). They do not depend on CYP2C9/VKORC1, so they are the preferred option in the PGx card.	FDA labels; genetic plan <i>moderate</i>	F5/F2: high CYP3A4: high
prasugrel Normal	CYP2B6 : RM	normal	CYP2B6 participates in metabolism/activation, without clinical recommendation by genotype. Usual dose.	PharmGKB (3 level); label FDA (prasugrel) <i>weak</i>	CYP2B6: moderate
rivaroxaban Normal	F5/F2 : non carrier CYP3A4 : normal	normal	Direct anticoagulants: without actionable PGx (the rivaroxaban label refers to F5 Leiden only in the context of thrombophilia; absent). They do not depend on CYP2C9/VKORC1, so they are the preferred option in the PGx card.	FDA labels; genetic plan <i>moderate</i>	F5/F2: high CYP3A4: high
ticagrelor Normal	CYP2C19 : NM CYP3A4 : normal	normal	It does not need activation by CYP2C19 (the label refers to it only by comparison with clopidogrel). Normal CYP3A4.	FDA label <i>moderate</i>	CYP2C19: high CYP3A4: high
ticlopidine Normal	CYP2B6 : RM	normal	CYP2B6 participates in metabolism/activation, without clinical recommendation by genotype. Usual dose.	PharmGKB (3 level); label FDA (prasugrel) <i>weak</i>	CYP2B6: moderate

Medication	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
acetylsalicylic acid Normal	G6PD : normal	normal	CPIC 2022 / labels: Normal G6PD → with no reason to avoid hemolysis risk. Rare variants of G6PD not excluded.	CPIC 2022 (G6PD); FDA labels <i>strong (CPIC A)</i>	G6PD: high for the tested variants
transexamic acid Normal	F5/F2 : non carrier	normal (without common genetic thrombophilia)	DPWG / labels: F5 Leiden and prothrombin G20210A increase the risk of thrombosis with estrogens and others. Both are absent → without additional genetic risk for these variants.	DPWG (F5 Leiden–estrogens); FDA labels <i>strong (for F5 Leiden)</i>	F5/F2: high

No known pharmacogenetic interaction (4):

dipyridamol (UGT)	fondaparinux (renal)
enoxaparin (renal)	unfractionated heparin (reticuloendothelial)

3.3 Analgesics, NSAIDs and opioids — 35 medicines

prevention: 2 · dose adjustment: 5 · Monitoring: 10 · normal: 9 · without known pharmacogenetic interaction: 9

Medication	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
piroxicam Avoid	CYP2C9 : IM (*1/*3)	avoid – choose alternative	CPIC 2020 (AS 1.0): choosing a NSAID not dependent on CYP2C9 or short half-life (very long half-lived → accumulation and risk GI/renal/CV).	CPIC 2020 <i>moderate/optimal (CPIC A; PharmGKB 1A)</i>	CYP2C9: high
tenoxicam Avoid	CYP2C9 : IM (*1/*3)	avoid – choose alternative	CPIC 2020 (AS 1.0): choosing a NSAID not dependent on CYP2C9 or short half-life (very long half-lived → accumulation and risk GI/renal/CV).	CPIC 2020 <i>moderate/optimal (CPIC A; PharmGKB 1A)</i>	CYP2C9: high
celecoxib Adjust dose	CYP2C9 : IM (*1/*3)	higher exposure → lower dose	CPIC 2020 (AS 1.0): start with the lowest recommended dose and holder with caution; course as short as possible. FDA label: *3 carriers with increased exposure.	CPIC 2020; FDA label <i>moderate (CPIC A; PharmGKB 1A)</i>	CYP2C9: high
flurbiprofen Adjust dose	CYP2C9 : IM (*1/*3)	higher exposure → lower dose	CPIC 2020 (AS 1.0): start with the lowest recommended dose, titer with caution, short course. Avoid chronic use. (Also sold without revenue.)	CPIC 2020; label FDA (flurbiprofen, ibuprofen) <i>moderate (CPIC A; PharmGKB 1A)</i>	CYP2C9: high
ibuprofen Adjust dose	CYP2C9 : IM (*1/*3)	higher exposure → lower dose	CPIC 2020 (AS 1.0): start with the lowest recommended dose, titer with caution, short course. Avoid chronic use. (Also sold without revenue.)	CPIC 2020; label FDA (flurbiprofen, ibuprofen) <i>moderate (CPIC A; PharmGKB 1A)</i>	CYP2C9: high
lornoxicam Adjust dose	CYP2C9 : IM (*1/*3)	higher exposure → lower dose	CPIC 2020 (AS 1.0): start with the lowest recommended dose, titer with caution, short course. Avoid chronic use. (Also sold without revenue.)	CPIC 2020; label FDA (flurbiprofen, ibuprofen) <i>moderate (CPIC A; PharmGKB 1A)</i>	CYP2C9: high
meloxicam Adjust dose	CYP2C9 : IM (*1/*3)	higher exposure → 50% of the dose	CPIC 2020 (AS 1.0): start with 50% of the lowest recommended dose and do not exceed 50% of the maximum dose, or choose alternative. Long half.	CPIC 2020; FDA label <i>moderate (CPIC A; PharmGKB 1A)</i>	CYP2C9: high
aceclofenac Monitoring	CYP2C9 : IM (*1/*3)	exposure possibly greater	CPIC 2020: no formal recommendation (insufficient evidence), but CYP2C9 is a relevant route. Use the lowest effective dose and the shortest course; avoid chronic use.	CPIC 2020 (without recommendation); PharmGKB <i>weak</i>	CYP2C9: high
codeine Monitoring	CYP2D6 : Probable NM (without CNV) SLC22A1 (OCT1) : normal	probable normal - confirm CYP2D6 with CNV	CPIC 2021 NM → label dose. The main risk (death due to excess morphine) is ultrafast, and a duplication of CYP2D6 is not visible in the chip. Before regular use, confirm CYP2D6 with clinical test with CNV. Normal OCT1.	CPIC 2021; DPWG; label FDA/EMA <i>strong (CPIC A; PharmGKB 1A)</i>	CYP2D6: moderate SLC22A1 (OCT1): high
diclofenac Monitoring	CYP2C9 : IM (*1/*3)	exposure possibly greater	CPIC 2020: no formal recommendation (insufficient evidence), but CYP2C9 is a relevant route. Use the lowest effective dose and the shortest course; avoid chronic use.	CPIC 2020 (without recommendation); PharmGKB <i>weak</i>	CYP2C9: high
indomethacin Monitoring	CYP2C9 : IM (*1/*3)	exposure possibly greater	CPIC 2020: no formal recommendation (insufficient evidence), but CYP2C9 is a relevant route. Use the lowest effective dose and the shortest course; avoid chronic use.	CPIC 2020 (without recommendation); PharmGKB <i>weak</i>	CYP2C9: high

Medication	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
methadone Monitoring	CYP2B6 : RM OPRM1 : A/G	possibly lower levels	Rapid CYP2B6 is associated with lower concentrations of methadone (PharmGKB 2A); CPIC without recommendation. Titular by response and ECG (QT).	CPIC (level C); PharmGKB <i>weak to moderate</i>	CYP2B6: moderate OPRM1: high
naproxen Monitoring	CYP2C9 : IM (*1/*3)	exposure possibly greater	CPIC 2020: no formal recommendation (insufficient evidence), but CYP2C9 is a relevant route. Use the lowest effective dose and the shortest course; avoid chronic use.	CPIC 2020 (without recommendation); PharmGKB <i>weak</i>	CYP2C9: high
nimesulide Monitoring	CYP2C9 : IM (*1/*3)	exposure possibly greater	CPIC 2020: no formal recommendation (insufficient evidence), but CYP2C9 is a relevant route. Use the lowest effective dose and the shortest course; avoid chronic use.	CPIC 2020 (without recommendation); PharmGKB <i>weak</i>	CYP2C9: high
paracetamol Monitoring	UGT1A1 : IM (*1/*28) PNPLA3/Pi*MZ (liver) : unfavourable	monitoring (liver)	No PGx guideline. UGT1A1*28 (some minor glucuronidation) and unfavourable liver genetics (PNPLA3, MBOAT7, Pi*MZ probable): the genetic plan suggests ≤2 g/day (suggestion, not applied).	Genetic plan; PharmGKB (3 level) <i>weak/indirect</i>	UGT1A1: high PNPLA3/Pi*MZ (liver): moderate
parecoxib Monitoring	CYP2C9 : IM (*1/*3)	exposure possibly greater	CPIC 2020: no formal recommendation (insufficient evidence), but CYP2C9 is a relevant route. Use the lowest effective dose and the shortest course; avoid chronic use.	CPIC 2020 (without recommendation); PharmGKB <i>weak</i>	CYP2C9: high
tramadol Monitoring	CYP2D6 : Probable NM (without CNV) SLC22A1 (OCT1) : normal	probable normal - confirm CYP2D6 with CNV	CPIC 2021 NM → label dose. Ultrafast (duplication, not visible in the chip) are at risk of toxicity (respiratory depression, serotonergic syndrome). Normal OCT1 (normal exposure to active metabolite). Confirm CYP2D6 with CNV before regular use.	CPIC 2021; DPWG; FDA label <i>strong (CPIC A; PharmGKB 1A)</i>	CYP2D6: moderate SLC22A1 (OCT1): high
buprenorphine Normal	OPRM1 : A/G COMT : Val/Val	normal	CPIC 2021 (opioides): no OPRM1 or COMT recommendation. The answer.	CPIC 2021 <i>strong (non-action)</i>	OPRM1: high COMT: high
fentanyl Normal	OPRM1 : A/G COMT : Val/Val	normal	CPIC 2021 (opioides): no OPRM1 or COMT recommendation. The answer.	CPIC 2021 <i>strong (non-action)</i>	OPRM1: high COMT: high
hydrocodone Normal	CYP2D6 : Probable NM (without CNV)	normal	The CYP2D6 is on the label/guidelines, but the recommendations apply to slow or ultrafast metabolizers. Probable NM → usual dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	FDA/EMA labels; CPIC 2021 (hydrocodone/oxycodone: no action for NM) <i>moderate</i>	CYP2D6: moderate
hydromorphone Normal	OPRM1 : A/G COMT : Val/Val	normal	CPIC 2021 (opioides): no OPRM1 or COMT recommendation. The answer.	CPIC 2021 <i>strong (non-action)</i>	OPRM1: high COMT: high
metamizole (dipyrone) Normal	NAT2 : slow	normal (minor effect)	The NAT2 acetyl metabolites; slow acetylators have different profiles of metabolites, with no established clinical relevance.	PharmGKB (3 level) <i>weak</i>	NAT2: high
morphine Normal	SLC22A1 (OCT1) : normal	normal	OCT1 of normal function (without the 4 variants of loss of function) → normal uptake/depuration. No guideline.	PharmGKB (2B–3 level) <i>weak</i>	SLC22A1 (OCT1): high
oxycodone Normal	CYP2D6 : Probable NM (without CNV)	normal	The CYP2D6 is on the label/guidelines, but the recommendations apply to slow or ultrafast metabolizers. Probable NM → usual dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	FDA/EMA labels; CPIC 2021 (hydrocodone/oxycodone: no action for NM) <i>moderate</i>	CYP2D6: moderate
tapentadol Normal	OPRM1 : A/G COMT : Val/Val	normal	CPIC 2021 (opioides): no OPRM1 or COMT recommendation. The answer.	CPIC 2021 <i>strong (non-action)</i>	OPRM1: high COMT: high

Medication	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
tizanidine Normal	CYP1A2: *1F het.	normal	CYP1A2*1F heterozygous: intermediate inductibility (the effect is noted mainly in smokers). No guideline; usual dose (clozapine and theophylline by levels, as always).	PharmGKB (3 level) weak	CYP1A2: high

No known pharmacogenetic interaction (9):

butalbital (CYP2C19) — CYP2C19: NM	cyclobenzaprine (CYP3A4; CYP1A2) — CYP3A4: normal; CYP1A2: *1F het.	metocarbamol (hepatic)
ketoprofen (UGT)	dexceptrufen (UGT)	nabumetone (CYP1A2) — CYP1A2: *1F het.
cetorolac (UGT; CYP2C8/2C9 (minor)) CYP2C8: normal	etoricoxib (CYP3A4) — CYP3A4: normal	naloxone (UGT)

3.4 Psychiatry — 64 medicines

Monitoring: 1 · normal: 38 · without known pharmacogenetic interaction: 25

Medication	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
pimozide Monitoring	CYP2D6 : Probable NM (without CNV)	probable normal – label requires genotyping above certain doses	FDA label: genotipar CYP2D6 before high doses (pimozide >4 mg/day; tetrabenazine >50 mg/day). With probable NM the usual titration is applied, but confirm with clinical test with CNV.	FDA label; CPIC (A/B) <i>strong</i>	CYP2D6: moderate
agomelatine Normal	CYP1A2 : *1F het.	normal	CYP1A2*1F heterozygous: intermediate inducibility (the effect is noted mainly in smokers). No guideline; usual dose (clozapine and theophylline by levels, as always).	PharmGKB (3 level) <i>weak</i>	CYP1A2: high
amitriptyline Normal	CYP2C19 : NM CYP2D6 : Probable NM (without CNV)	normal	CPIC 2016/2019: CYP2C19 NM + CYP2D6 NM probable → recommended initial dose; serum levels were high. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	CPIC (tricyclic); DPWG <i>strong</i> (CPIC A/B; PharmGKB 1A)	CYP2C19: high CYP2D6: moderate
amphetamine/ dextroamphetamine Normal	CYP2D6 : Probable NM (without CNV)	normal	The CYP2D6 is on the label/guidelines, but the recommendations apply to slow or ultrafast metabolizers. Probable NM → usual dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	FDA/EMA labels; CPIC 2021 (hydrocodone/oxycodone: no action for NM) <i>moderate</i>	CYP2D6: moderate
aripiprazole Normal	CYP2D6 : Probable NM (without CNV)	normal	DPWG / FDA labels: adjustments only for slow or ultrafast metabolizers. Probable NM → usual dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	DPWG; FDA labels; CPIC (B/C) <i>moderate</i> (PharmGKB 1A for aripiprazole/ haloperidol/risperidone)	CYP2D6: moderate
atomoxetine Normal	CYP2D6 : Probable NM (without CNV)	normal	CPIC 2019 (adults NM): start 40 mg/day, rise to 80 mg/day after 3 days; without response to 2 weeks, until 100 mg/day. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	CPIC 2019; DPWG; FDA label <i>moderate</i> (CPIC A)	CYP2D6: moderate
brexpiprazole Normal	CYP2D6 : Probable NM (without CNV)	normal	DPWG / FDA labels: adjustments only for slow or ultrafast metabolizers. Probable NM → usual dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	DPWG; FDA labels; CPIC (B/C) <i>moderate</i> (PharmGKB 1A for aripiprazole/ haloperidol/risperidone)	CYP2D6: moderate
bupropion Normal	CYP2B6 : RM	normal (minor effect)	CPIC (B/C): without recommendation; CYP2B6*4 tends to form more hydroxybupropion. Usual dose.	CPIC (B/C); PharmGKB 2A <i>weak</i>	CYP2B6: moderate
cariprazine Normal	CYP2D6 : Probable NM (without CNV)	normal	DPWG / FDA labels: adjustments only for slow or ultrafast metabolizers. Probable NM → usual dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	DPWG; FDA labels; CPIC (B/C) <i>moderate</i> (PharmGKB 1A for aripiprazole/ haloperidol/risperidone)	CYP2D6: moderate
citalopram Normal	CYP2C19 : NM	normal	CPIC 2023: normal metabolizer → recommended initial dose.	CPIC 2023; DPWG; FDA label <i>strong</i> (CPIC A; PharmGKB 1A)	CYP2C19: high

Medication	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
clomipramine Normal	CYP2C19 : NM CYP2D6 : Probable NM (without CNV)	normal	CPIC 2016/2019: CYP2C19 NM + CYP2D6 NM probable → recommended initial dose; serum levels were high. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	CPIC (tricyclic); DPWG <i>strong</i> (CPIC A/B; PharmGKB 1A)	CYP2C19: high CYP2D6: moderate
clonazepam Normal	NAT2 : slow	normal (minor effect)	The NAT2 acetyl metabolites; slow acetylators have different profiles of metabolites, with no established clinical relevance.	PharmGKB (3 level) <i>weak</i>	NAT2: high
clozapine Normal	CYP1A2 : *1F het.	normal	CYP1A2*1F heterozygous: intermediate inducibility (the effect is noted mainly in smokers). No guideline; usual dose (clozapine and theophylline by levels, as always).	PharmGKB (3 level) <i>weak</i>	CYP1A2: high
desvenlafaxine Normal	CYP2D6 : Probable NM (without CNV)	normal	CPIC 2023/DPWG NM → recommended initial dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	CPIC 2023; DPWG; labels FDA <i>strong for vortioxetine/ paroxetine/venlafaxine</i> (CPIC A/B); remaining <i>informative</i>	CYP2D6: moderate
diazepam Normal	CYP2C19 : NM	normal	The CYP2C19 is a relevant route and is included in the label/guidelines, but the recommendations apply to intermediate/slow metabolizers. Normal metabolizer → usual dose.	Labels FDA/EMA; CPIC (level B/C) <i>moderate</i>	CYP2C19: high
doxepin Normal	CYP2C19 : NM CYP2D6 : Probable NM (without CNV)	normal	CPIC 2016/2019: CYP2C19 NM + CYP2D6 NM probable → recommended initial dose; serum levels were high. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	CPIC (tricyclic); DPWG <i>strong</i> (CPIC A/B; PharmGKB 1A)	CYP2C19: high CYP2D6: moderate
duloxetine Normal	CYP2D6 : Probable NM (without CNV)	normal	CPIC 2023/DPWG NM → recommended initial dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	CPIC 2023; DPWG; labels FDA <i>strong for vortioxetine/ paroxetine/venlafaxine</i> (CPIC A/B); remaining <i>informative</i>	CYP2D6: moderate
escitalopram Normal	CYP2C19 : NM	normal	CPIC 2023: normal metabolizer → recommended initial dose.	CPIC 2023; DPWG; FDA label <i>strong</i> (CPIC A; PharmGKB 1A)	CYP2C19: high
fluoxetine Normal	CYP2D6 : Probable NM (without CNV)	normal	CPIC 2023/DPWG NM → recommended initial dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	CPIC 2023; DPWG; labels FDA <i>strong for vortioxetine/ paroxetine/venlafaxine</i> (CPIC A/B); remaining <i>informative</i>	CYP2D6: moderate
fluvoxamine Normal	CYP2D6 : Probable NM (without CNV)	normal	CPIC 2023/DPWG NM → recommended initial dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	CPIC 2023; DPWG; labels FDA <i>strong for vortioxetine/ paroxetine/venlafaxine</i> (CPIC A/B); remaining <i>informative</i>	CYP2D6: moderate

Medication	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
haloperidol Normal	CYP2D6 : Probable NM (without CNV)	normal	DPWG / FDA labels: adjustments only for slow or ultrafast metabolizers. Probable NM → usual dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2-5%), ultrafast.	DPWG; FDA labels; CPIC (B/C) <i>moderate (PharmGKB 1A for aripiprazole/ haloperidol/risperidone)</i>	CYP2D6: moderate
imipramine Normal	CYP2C19 : NM CYP2D6 : Probable NM (without CNV)	normal	CPIC 2016/2019: CYP2C19 NM + CYP2D6 NM probable → recommended initial dose; serum levels were high. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2-5%), ultrafast.	CPIC (tricyclic); DPWG <i>strong (CPIC A/B; PharmGKB 1A)</i>	CYP2C19: high CYP2D6: moderate
lisdexamphetamine Normal	CYP2D6 : Probable NM (without CNV)	normal	The CYP2D6 is on the label/guidelines, but the recommendations apply to slow or ultrafast metabolizers. Probable NM → usual dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2-5%), ultrafast.	FDA/EMA labels; CPIC 2021 (hydrocodone/oxycodone: no action for NM) <i>moderate</i>	CYP2D6: moderate
melatonin Normal	CYP1A2 : *1F het.	normal	CYP1A2*1F heterozygous: intermediate inducibility (the effect is noted mainly in smokers). No guideline; usual dose (clozapine and theophylline by levels, as always).	PharmGKB (3 level) <i>weak</i>	CYP1A2: high
mirtazapine Normal	CYP2D6 : Probable NM (without CNV)	normal	CPIC 2023/DPWG NM → recommended initial dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2-5%), ultrafast.	CPIC 2023; DPWG; labels FDA <i>strong for vortioxetine/ paroxetine/venlafaxine (CPIC A/B); remaining informative</i>	CYP2D6: moderate
modafinil Normal	CYP2D6 : Probable NM (without CNV)	normal	The CYP2D6 is on the label/guidelines, but the recommendations apply to slow or ultrafast metabolizers. Probable NM → usual dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2-5%), ultrafast.	FDA/EMA labels; CPIC 2021 (hydrocodone/oxycodone: no action for NM) <i>moderate</i>	CYP2D6: moderate
★ naltrexone (incl. low dose) Normal	OPRM1 : A/G	normal	OPRM1 118 A/G: some studies (alcoholism) give better response to carriers G; inconsistent evidence. CPIC without recommendation.	PharmGKB (2B-3 level) <i>weak</i>	OPRM1: high
nortriptyline Normal	CYP2D6 : Probable NM (without CNV)	normal	CPIC 2023/DPWG NM → recommended initial dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2-5%), ultrafast.	CPIC 2023; DPWG; labels FDA <i>strong for vortioxetine/ paroxetine/venlafaxine (CPIC A/B); remaining informative</i>	CYP2D6: moderate
olanzapine Normal	CYP1A2 : *1F het.	normal	CYP1A2*1F heterozygous: intermediate inducibility (the effect is noted mainly in smokers). No guideline; usual dose (clozapine and theophylline by levels, as always).	PharmGKB (3 level) <i>weak</i>	CYP1A2: high
paliperidone Normal	CYP2D6 : Probable NM (without CNV)	normal	The CYP2D6 is on the label, without dose adjustment for normal metabolizers. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2-5%), ultrafast.	FDA labels <i>moderate</i>	CYP2D6: moderate

Medication	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
paroxetine Normal	CYP2D6 : Probable NM (without CNV)	normal	CPIC 2023/DPWG NM → recommended initial dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	CPIC 2023; DPWG; labels FDA <i>strong for vortioxetine/paroxetine/venlafaxine (CPIC A/B); remaining informative</i>	CYP2D6: moderate
perfenazine Normal	CYP2D6 : Probable NM (without CNV)	normal	DPWG / FDA labels: adjustments only for slow or ultrafast metabolizers. Probable NM → usual dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	DPWG; FDA labels; CPIC (B/C) <i>moderate (PharmGKB 1A for aripiprazole/haloperidol/risperidone)</i>	CYP2D6: moderate
risperidone Normal	CYP2D6 : Probable NM (without CNV)	normal	DPWG / FDA labels: adjustments only for slow or ultrafast metabolizers. Probable NM → usual dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	DPWG; FDA labels; CPIC (B/C) <i>moderate (PharmGKB 1A for aripiprazole/haloperidol/risperidone)</i>	CYP2D6: moderate
sertraline Normal	CYP2C19 : NM CYP2B6 : RM	normal	CPIC 2023 CYP2C19 NM + CYP2B6 fast → recommended initial dose (the rapid CYP2B6 can slightly lower the levels; without action).	CPIC 2023 <i>moderate (CPIC A/B)</i>	CYP2C19: high CYP2B6: moderate
trimipramine Normal	CYP2C19 : NM CYP2D6 : Probable NM (without CNV)	normal	CPIC 2016/2019: CYP2C19 NM + CYP2D6 NM probable → recommended initial dose; serum levels were high. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	CPIC (tricyclic); DPWG <i>strong (CPIC A/B; PharmGKB 1A)</i>	CYP2C19: high CYP2D6: moderate
venlafaxine Normal	CYP2D6 : Probable NM (without CNV)	normal	CPIC 2023/DPWG NM → recommended initial dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	CPIC 2023; DPWG; labels FDA <i>strong for vortioxetine/paroxetine/venlafaxine (CPIC A/B); remaining informative</i>	CYP2D6: moderate
viloxazine Normal	CYP2D6 : Probable NM (without CNV)	normal	The CYP2D6 is on the label/guidelines, but the recommendations apply to slow or ultrafast metabolizers. Probable NM → usual dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	FDA/EMA labels; CPIC 2021 (hydrocodone/oxycodone: no action for NM) <i>moderate</i>	CYP2D6: moderate
vortioxetine Normal	CYP2D6 : Probable NM (without CNV)	normal	CPIC 2023/DPWG NM → recommended initial dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	CPIC 2023; DPWG; labels FDA <i>strong for vortioxetine/paroxetine/venlafaxine (CPIC A/B); remaining informative</i>	CYP2D6: moderate
zuclopentixol Normal	CYP2D6 : Probable NM (without CNV)	normal	DPWG / FDA labels: adjustments only for slow or ultrafast metabolizers. Probable NM → usual dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	DPWG; FDA labels; CPIC (B/C) <i>moderate (PharmGKB 1A for aripiprazole/haloperidol/risperidone)</i>	CYP2D6: moderate

No known pharmacogenetic interaction (25):

acamprosate (renal)

alprazolam (CYP3A4) — CYP3A4: normal

amissulprid (renal)

bromazepam (CYP3A4) — *CYP3A4: normal*

buspirone (CYP3A4) — *CYP3A4: normal*

chlorpromazine (CYP2D6; CYP1A2) — *CYP2D6: Probable NM (without CNV); CYP1A2: *1F het.*

dexmethylphenidate (CES1)

disulfiram (hepatic)

guanfacine (CYP3A4) — *CYP3A4: normal*

ethyl loflazepat (hepatic)

lorazepam (UGT2B15)

lurasidone (CYP3A4) — *CYP3A4: normal*

lithium (renal)

methylphenidate (CES1)

myanserin (CYP2D6; CYP1A2) — *CYP2D6: Probable NM (without CNV); CYP1A2: *1F het.*

nicotine (substitution) (CYP2A6) — *CYP2A6: probable normal*

oxazepam (UGT2B15)

quetiapine (CYP3A4) — *CYP3A4: normal*

sulpiride (renal)

temazepam (UGT)

trazodone (CYP3A4) — *CYP3A4: normal*

varenicline (renal)

ziprasidone (aldehyde oxidase; CYP3A4) *CYP3A4: normal*

zolepidem (CYP3A4) — *CYP3A4: normal*

zopiclone (CYP3A4; CYP2C8) — *CYP3A4: normal; CYP2C8: normal*

3.5 Neurology — 53 medicines

dose adjustment: 4 · Monitoring: 2 · normal: 16 · without known pharmacogenetic interaction: 31

Medication	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
amifampridin Adjust dose	NAT2 : slow	higher exposure → lower initial dose	FDA label: slow metabolizers NAT2 → start with the lowest recommended daily dose.	FDA label <i>strong</i>	NAT2: high
phenytoin Adjust dose	CYP2C9 : IM (*1/*3) HLA-B*15:02 : n/a (very low priority)	higher exposure → ~25% less maintenance	CPIC2020: 1.the usual dose/load; maintenance ~25% below the usual, adjusted for serum levels. HLA-B*15:02 not evaluated, but unlikely in the reference population.	CPIC 2020; DPWG; FDA label <i>moderate/strong (CPIC A; PharmGKB 1A)</i>	CYP2C9: high HLA-B*15:02: non-determinable
phosphenitoin Adjust dose	CYP2C9 : IM (*1/*3) HLA-B*15:02 : n/a (very low priority)	higher exposure → ~25% less maintenance	CPIC2020: 1.the usual dose/load; maintenance ~25% below the usual, adjusted for serum levels. HLA-B*15:02 not evaluated, but unlikely in the reference population.	CPIC 2020; DPWG; FDA label <i>moderate/strong (CPIC A; PharmGKB 1A)</i>	CYP2C9: high HLA-B*15:02: non-determinable
siponimod Adjust dose	CYP2C9 : IM (*1/*3)	dose adjustment (maintenance 1 mg)	FDA/EMA labels: mandatory CYP2C9 genotyping. *1/*3 → maintenance dose 1 mg/day (instead of 2 mg); *3/*3 contraindicated.	Label FDA/EMA; CPIC (level A) <i>strong (compulsory test)</i>	CYP2C9: high
deutetrabenazine Monitoring	CYP2D6 : Probable NM (without CNV)	probable normal – label requires genotyping above certain doses	FDA label: genotipar CYP2D6 before high doses (pimozide >4 mg/day; tetrabenazine >50 mg/day). With probable NM the usual titration is applied, but confirm with clinical test with CNV.	FDA label; CPIC (A/B) <i>strong</i>	CYP2D6: moderate
tetrabenazine Monitoring	CYP2D6 : Probable NM (without CNV)	probable normal – label requires genotyping above certain doses	FDA label: genotipar CYP2D6 before high doses (pimozide >4 mg/day; tetrabenazine >50 mg/day). With probable NM the usual titration is applied, but confirm with clinical test with CNV.	FDA label; CPIC (A/B) <i>strong</i>	CYP2D6: moderate
brivalacetam Normal	CYP2C19 : NM	normal	The CYP2C19 is a relevant route and is included in the label/guidelines, but the recommendations apply to intermediate/slow metabolizers. Normal metabolizer → usual dose.	Labels FDA/EMA; CPIC (level B/C) <i>moderate</i>	CYP2C19: high
carbamazepine Normal	HLA-A*31:01 : probable negative HLA-B*15:02 : n/a (very low priority)	probable normal	CPIC 2017 A*31:01 and B*15:02 negative doses. A*31:01 (≈2–5%) in the reference population; DRESS/SJS) is probable negative by imputed tag; B*15:02 almost nonexistent in the reference population. Confirm A*31:01 by typing if it is long therapy.	CPIC 2017; DPWG; FDA label <i>strong (CPIC A; PharmGKB 1A)</i>	HLA-A*31:01: moderate HLA-B*15:02: non-determinable
clobazam Normal	CYP2C19 : NM	normal	The CYP2C19 is a relevant route and is included in the label/guidelines, but the recommendations apply to intermediate/slow metabolizers. Normal metabolizer → usual dose.	Labels FDA/EMA; CPIC (level B/C) <i>moderate</i>	CYP2C19: high
donepezil Normal	CYP2D6 : Probable NM (without CNV)	normal	The CYP2D6 is on the label/guidelines, but the recommendations apply to slow or ultrafast metabolizers. Probable NM → usual dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	FDA/EMA labels; CPIC 2021 (hydrocodone/oxycodone: no action for NM) <i>moderate</i>	CYP2D6: moderate
slicarbazepine Normal	HLA-B*15:02 : n/a (very low priority)	normal	CPIC 2017/labels: the risk of SJS/NET binds to HLA-B*15:02. Not evaluated, but with very low a priori probability in the reference population.	CPIC 2017; FDA labels <i>strong for B*15:02 (applicable mainly where the allele is frequent)</i>	HLA-B*15:02: non-determinable

Medication	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
galantamine Normal	CYP2D6 : Probable NM (without CNV)	normal	The CYP2D6 is on the label/guidelines, but the recommendations apply to slow or ultrafast metabolizers. Probable NM → usual dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	FDA/EMA labels; CPIC 2021 (hydrocodone/oxycodone: no action for NM) <i>moderate</i>	CYP2D6: moderate
lacosamide Normal	CYP2C19 : NM	normal	The CYP2C19 is a relevant route and is included in the label/guidelines, but the recommendations apply to intermediate/slow metabolizers. Normal metabolizer → usual dose.	Labels FDA/EMA; CPIC (level B/C) <i>moderate</i>	CYP2C19: high
lamotrigine Normal	HLA-B*15:02 : n/a (very low priority)	normal	CPIC 2017/labels: the risk of SJS/NET binds to HLA-B*15:02. Not evaluated, but with very low a priori probability in the reference population.	CPIC 2017; FDA labels <i>strong for B*15:02 (applicable mainly where the allele is frequent)</i>	HLA-B*15:02: non-determinable
oxcarbazepine Normal	HLA-B*15:02 : n/a (very low priority)	normal	CPIC 2017/labels: the risk of SJS/NET binds to HLA-B*15:02. Not evaluated, but with very low a priori probability in the reference population.	CPIC 2017; FDA labels <i>strong for B*15:02 (applicable mainly where the allele is frequent)</i>	HLA-B*15:02: non-determinable
pitolisant Normal	CYP2D6 : Probable NM (without CNV)	normal	The CYP2D6 is on the label/guidelines, but the recommendations apply to slow or ultrafast metabolizers. Probable NM → usual dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	FDA/EMA labels; CPIC 2021 (hydrocodone/oxycodone: no action for NM) <i>moderate</i>	CYP2D6: moderate
riluzole Normal	CYP1A2 : *1F het.	normal	CYP1A2*1F heterozygous: intermediate inducibility (the effect is noted mainly in smokers). No guideline; usual dose (clozapine and theophylline by levels, as always).	PharmGKB (3 level) <i>weak</i>	CYP1A2: high
rimegepant Normal	CYP2C9 : IM (*1/*3)	normal	CYP2C9 contributes to metabolism, but label recommendations apply only to slow metabolizers (*3/*3) or associations with inhibitors. Usual dose.	FDA/EMA labels <i>moderate</i>	CYP2C9: high
ropinirol Normal	CYP1A2 : *1F het.	normal	CYP1A2*1F heterozygous: intermediate inducibility (the effect is noted mainly in smokers). No guideline; usual dose (clozapine and theophylline by levels, as always).	PharmGKB (3 level) <i>weak</i>	CYP1A2: high
sumatriptane Normal	SLC22A1 (OCT1) : normal	normal	OCT1 of normal function (without the 4 variants of loss of function) → normal uptake/depuration. No guideline.	PharmGKB (2B–3 level) <i>weak</i>	SLC22A1 (OCT1): high
valbenazine Normal	CYP2D6 : Probable NM (without CNV)	normal	The CYP2D6 is on the label/guidelines, but the recommendations apply to slow or ultrafast metabolizers. Probable NM → usual dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	FDA/EMA labels; CPIC 2021 (hydrocodone/oxycodone: no action for NM) <i>moderate</i>	CYP2D6: moderate
valproic acid/ valproate Normal	POLG : normal CYP2C9 : IM (*1/*3)	normal	FDA / CPIC label: contraindicated in mitochondrial disease by POLG. Common POLG variants (A467T, W748S) are absent. CYP2C9 is minor. Hepatic surveillance.	FDA label; CPIC (A/B) <i>strong (POLG)</i>	POLG: high for these variants CYP2C9: high

No known pharmacogenetic interaction (31):

amantadine (renal)
baclofen (renal)
beta-histin (MAO)
cenobamate (UGT2B7; CYP2E1/2A6)
entacapone (UGT)

erenumab (proteolysis)
etosuximide (CYP3A4) — *CYP3A4: normal*
phenobarbital (CYP2C9; CYP2C19) — *CYP2C9: IM (*1/*3); CYP2C19: NM*
fingolimod (CYP4F2) — *CYP4F2: normal*

flunarizine (CYP2D6) — *CYP2D6: Probable NM (without CNV)*
Dimethyl fumarate (sterases)
gabapentin (renal)
levetiracetam (hydrolysis; renal)
levodopa / carbidopa (DDC; COMT)

memantine (renal)
ocrelizumab (proteolysis)
opicapone (sulfation)
ozanimod (MAO-B; CYP2C8) — *CYP2C8: normal*
perampanel (CYP3A4) — *CYP3A4: normal*
pramipexole (renal)

pregabalin (renal)
primidone (CYP2C9/2C19 (→ phenobarbital)) *CYP2C9: IM (*1/*3)*
rasagiline (CYP1A2) — *CYP1A2: *1F het.*
rivastigmine (cholinesterases)
rhizatriptan (MAO-A)
safinamide (amideses)

teriflunomide (hydrolysis; BCRP) *ABCG2: normal*
topiramate (renal)
ubigepant (CYP3A4) — *CYP3A4: normal*
zolmitriptan (CYP1A2; MAO-A) — *CYP1A2: *1F het.*
zonisamide (CYP3A4) — *CYP3A4: normal*

3.6 Anti-infectives — 85 medicines

dose adjustment: 1 · Monitoring: 2 · indeterminate: 6 · normal: 27 · without known pharmacogenetic interaction: 49

Medication	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
isoniazid Adjust dose	NAT2 : slow PNPLA3/Pi*MZ (liver) : unfavourable	higher exposure → hepatotoxicity	Slow acetylators have higher levels and more hepatotoxicity and neuropathy (FDA label; PharmGKB 1B). NAT2-guided schemas use a lower dose in slow ones. Add pyridoxin and tight hepatic surveillance (there are also unfavourable liver genetics).	Label FDA; PharmGKB 1B; CPIC (level C) <i>moderate to strong</i>	NAT2: high PNPLA3/Pi*MZ (liver): moderate
dapsone Monitoring	NAT2 : slow G6PD : normal	more hematological toxicity possible	Normal G6PD → with no reason to avoid (CPIC 2022). Slow acetylator: higher levels and more adverse effects (metemoglobinemia) described. HLA-B*13:01 not evaluated.	CPIC 2022 (G6PD); PharmGKB <i>moderate</i>	NAT2: high G6PD: high for the tested variants
sulfamethoxazole (with trimethoprim) Monitoring	NAT2 : slow G6PD : normal	more hypersensitivity possible	Slow acetylators have more hypersensitivity reactions to sulfonamides (FDA label NAT); PharmGKB). Normal G6PD → without restriction of hemolysis (CPIC 2022).	Label FDA; CPIC 2022 (G6PD) <i>moderate</i>	NAT2: high G6PD: high for the tested variants
amikacin Undetermined	MT-RNR1 : indeterminate	indeterminate (mitochondrial ototoxicity)	CPIC 2021: carriers of m.1555A>G/m.1494C>T should avoid aminoglycosides without any alternative. The MT-RNR1 state is not determinable in this chip. Without result, CPIC indicates normal use, short course and auditory surveillance; if repeated or prolonged use is predicted, clinical testing of m.1555A>G (suggestion, not applied). Oral/topic neomycin has low absorption.	CPIC 2021; FDA labels <i>strong (CPIC A; PharmGKB 1A)</i>	MT-RNR1: non-determinable
streptomycin Undetermined	MT-RNR1 : indeterminate	indeterminate (mitochondrial ototoxicity)	CPIC 2021: carriers of m.1555A>G/m.1494C>T should avoid aminoglycosides without any alternative. The MT-RNR1 state is not determinable in this chip. Without result, CPIC indicates normal use, short course and auditory surveillance; if repeated or prolonged use is predicted, clinical testing of m.1555A>G (suggestion, not applied). Oral/topic neomycin has low absorption.	CPIC 2021; FDA labels <i>strong (CPIC A; PharmGKB 1A)</i>	MT-RNR1: non-determinable
gentamicin Undetermined	MT-RNR1 : indeterminate	indeterminate (mitochondrial ototoxicity)	CPIC 2021: carriers of m.1555A>G/m.1494C>T should avoid aminoglycosides without any alternative. The MT-RNR1 state is not determinable in this chip. Without result, CPIC indicates normal use, short course and auditory surveillance; if repeated or prolonged use is predicted, clinical testing of m.1555A>G (suggestion, not applied). Oral/topic neomycin has low absorption.	CPIC 2021; FDA labels <i>strong (CPIC A; PharmGKB 1A)</i>	MT-RNR1: non-determinable
neomycin Undetermined	MT-RNR1 : indeterminate	indeterminate (mitochondrial ototoxicity)	CPIC 2021: carriers of m.1555A>G/m.1494C>T should avoid aminoglycosides without any alternative. The MT-RNR1 state is not determinable in this chip. Without result, CPIC indicates normal use, short course and auditory surveillance; if repeated or prolonged use is predicted, clinical testing of m.1555A>G (suggestion, not applied). Oral/topic neomycin has low absorption.	CPIC 2021; FDA labels <i>strong (CPIC A; PharmGKB 1A)</i>	MT-RNR1: non-determinable
terbinafine Undetermined	HLA-A*33:01 : indeterminate PNPLA3/Pi*MZ (liver) : unfavourable	indeterminate (liver risk)	HLA-A*33:01 is associated with liver injury by oral terbinafine (PharmGKB) and is not evaluated here. Unfavourable liver genetic context (PNPLA3, MBOAT7, probable Pi*MZ): is not a PGx guideline, but justifies surveillance of transaminases. To prefer topical treatment when possible (suggestion, not applied).	PharmGKB; DILI studies <i>moderate</i>	HLA-A*33:01: non-determinable PNPLA3/Pi*MZ (liver): moderate

Medication	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
tobramycin Undetermined	MT-RNR1: indeterminate	indeterminate (mitochondrial ototoxicity)	CPIC 2021: carriers of m.1555A>G/m.1494C>T should avoid aminoglycosides without any alternative. The MT-RNR1 state is not determinable in this chip. Without result, CPIC indicates normal use, short course and auditory surveillance; if repeated or prolonged use is predicted, clinical testing of m.1555A>G (suggestion, not applied). Oral/topic neomycin has low absorption.	CPIC 2021; FDA labels <i>strong (CPIC A; PharmGKB 1A)</i>	MT-RNR1: non-determinable
abacavir Normal	HLA-B*57:01: negative	normal (non-augmented hypersensitivity risk)	CPIC 2014 B*57:01 negative → use in the standard dose. The label requires clinical HLA typing before starting (the chip does not replace it).	CPIC 2014; DPWG; label FDA/EMA (compulsory test) <i>strong (CPIC A; PharmGKB 1A)</i>	HLA-B*57:01
atazanavir Normal	UGT1A1: IM (*1/*28)	normal (unprobable jaundice)	CPIC 2015: *1/*28 → without need to avoid; low probability of jaundice suspension.	CPIC 2015 <i>strong (CPIC A)</i>	UGT1A1: high
atovaquone + proguanil Normal	CYP2C19: NM	normal	The CYP2C19 is a relevant route and is included in the label/guidelines, but the recommendations apply to intermediate/slow metabolizers. Normal metabolizer → usual dose.	Labels FDA/EMA; CPIC (level B/C) <i>moderate</i>	CYP2C19: high
bictegravir Normal	UGT1A1: IM (*1/*28)	normal	Glucuronidated by UGT1A1; the labels mention only *28/*28 (mild exposure increase, without adjustment). Usual dose.	FDA labels <i>moderate</i>	UGT1A1: high
cabotegravir Normal	UGT1A1: IM (*1/*28)	normal	Glucuronidated by UGT1A1; the labels mention only *28/*28 (mild exposure increase, without adjustment). Usual dose.	FDA labels <i>moderate</i>	UGT1A1: high
ceftriaxone Normal	G6PD: normal	normal	CPIC 2022 / labels: Normal G6PD → with no reason to avoid hemolysis risk. Rare variants of G6PD not excluded.	CPIC 2022 (G6PD); FDA labels <i>strong (CPIC A)</i>	G6PD: high for the tested variants
ciprofloxacin Normal	G6PD: normal	normal	CPIC 2022 / labels: Normal G6PD → with no reason to avoid hemolysis risk. Rare variants of G6PD not excluded.	CPIC 2022 (G6PD); FDA labels <i>strong (CPIC A)</i>	G6PD: high for the tested variants
chloroquine Normal	G6PD: normal	normal	CPIC 2022 / labels: Normal G6PD → with no reason to avoid hemolysis risk. Rare variants of G6PD not excluded.	CPIC 2022 (G6PD); FDA labels <i>strong (CPIC A)</i>	G6PD: high for the tested variants
dolutegravir Normal	UGT1A1: IM (*1/*28)	normal	Glucuronidated by UGT1A1; the labels mention only *28/*28 (mild exposure increase, without adjustment). Usual dose.	FDA labels <i>moderate</i>	UGT1A1: high
efavirenz Normal	CYP2B6: RM	normal	CPIC 2019: rapid metabolizer → standard dose (600 mg/day).	CPIC 2019; DPWG <i>strong (CPIC A; PharmGKB 1A)</i>	CYP2B6: moderate
flucytosine Normal	DPYD: normal (AS 2)	normal	CPIC 2023/DPWG: DPYD AS 2 → label dose. The 13 variants tested include the reference 4 (CPIC/EMA). Before chemotherapy, EMA recommends clinical DPD test; repeat because rare variants are not seen in the chip. (It also applies to the topic 5-FU.)	CPIC 2023; DPWG; EMA 2020; FDA label <i>strong (CPIC A; PharmGKB 1A)</i>	DPYD: high for the tested variants
flucloxacillin Normal	HLA-B*57:01: negative	normal	HLA-B*57:01 is associated with hepatic injury by flucloxacillin (PharmGKB 1B-2A). Negative → not increased risk.	PharmGKB; DPWG (no action) <i>moderate</i>	HLA-B*57:01

Medication	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
glecaprevir + pibrentesvir Normal	IFNL3/IFNL4: favourable (CC)	normal	Direct antivirals: IFNL3 is no longer clinically relevant.	FDA labels (historical IFNL3) <i>weak</i>	IFNL3/IFNL4: high
levofloxacin Normal	G6PD: normal	normal	CPIC 2022 / labels: Normal G6PD → with no reason to avoid hemolysis risk. Rare variants of G6PD not excluded.	CPIC 2022 (G6PD); FDA labels <i>strong (CPIC A)</i>	G6PD: high for the tested variants
moxifloxacin Normal	G6PD: normal	normal	CPIC 2022 / labels: Normal G6PD → with no reason to avoid hemolysis risk. Rare variants of G6PD not excluded.	CPIC 2022 (G6PD); FDA labels <i>strong (CPIC A)</i>	G6PD: high for the tested variants
nevirapine Normal	CYP2B6: RM HLA-B*57:01: negative	normal	Rapid CYP2B6: normal/higher clearance; no adjustment recommendation.	CPIC (B/C); PharmGKB <i>weak</i>	CYP2B6: moderate HLA-B*57:01
nitrofurantoin Normal	G6PD: normal	normal	CPIC 2022 / labels: Normal G6PD → with no reason to avoid hemolysis risk. Rare variants of G6PD not excluded.	CPIC 2022 (G6PD); FDA labels <i>strong (CPIC A)</i>	G6PD: high for the tested variants
ofloxacin Normal	G6PD: normal	normal	CPIC 2022 / labels: Normal G6PD → with no reason to avoid hemolysis risk. Rare variants of G6PD not excluded.	CPIC 2022 (G6PD); FDA labels <i>strong (CPIC A)</i>	G6PD: high for the tested variants
peginterferon alfa Normal	IFNL3/IFNL4: favourable (CC)	favourable response	IFNL3 rs12979860 C/C (and IFNL4 TT/TT): a higher probability of sustained virological response to interferon therapy (hepatitis C). Guideline CPIC removed because these regimens are no longer used.	CPIC 2014 (withdrawal); PharmGKB 1B <i>moderate</i>	IFNL3/IFNL4: high
primaquine Normal	CYP2D6: Probable NM (without CNV) G6PD: normal	normal	Normal G6PD → without restriction of hemolysis (CPIC 2022). CYP2D6NM probable → normal activation (slow metabolizers have therapeutic failure in vivax malaria).	CPIC 2022 (G6PD); label FDA <i>strong (G6PD)</i>	CYP2D6: moderate G6PD: high for the tested variants
quinine Normal	G6PD: normal	normal	CPIC 2022 / labels: Normal G6PD → with no reason to avoid hemolysis risk. Rare variants of G6PD not excluded.	CPIC 2022 (G6PD); FDA labels <i>strong (CPIC A)</i>	G6PD: high for the tested variants
raltegravir Normal	UGT1A1: IM (*1/*28)	normal	Glucuronidated by UGT1A1; the labels mention only *28/*28 (mild exposure increase, without adjustment). Usual dose.	FDA labels <i>moderate</i>	UGT1A1: high
ribavirin Normal	IFNL3/IFNL4: favourable (CC)	favourable response	IFNL3 rs12979860 C/C (and IFNL4 TT/TT): a higher probability of sustained virological response to interferon therapy (hepatitis C). Guideline CPIC removed because these regimens are no longer used.	CPIC 2014 (withdrawal); PharmGKB 1B <i>moderate</i>	IFNL3/IFNL4: high
sofosbuvir + velpatasvir Normal	IFNL3/IFNL4: favourable (CC)	normal	Direct antivirals: IFNL3 is no longer clinically relevant.	FDA labels (historical IFNL3) <i>weak</i>	IFNL3/IFNL4: high
tapheñoquinone Normal	G6PD: normal CYP2D6: Probable NM (without CNV)	normal	CPIC 2022 Normal G6PD → with no reason to avoid. (Quantimate G6PD test remains mandatory by the label.)	CPIC 2022; FDA label <i>strong</i>	G6PD: high for the tested variants CYP2D6: moderate
voriconazole Normal	CYP2C19: NM	normal	CPIC 2016: normal metabolizer → standard dose (with monitoring of levels, as usual).	CPIC 2016; DPWG; FDA label <i>strong (CPIC A; PharmGKB 1A)</i>	CYP2C19: high

Medication	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
clavulanic acid (with amoxicillin) Normal	HLA-DRB1*15:01: negative	normal	Hepatic injury by amoxicillin-clavulanate is associated with HLA-DRB1*15:01 (PharmGKB 2B). Negative.	PharmGKB <i>weak to moderate</i>	HLA-DRB1*15:01: moderate

No known pharmacogenetic interaction (49):

acyclovir (renal)	entecavir (renal)	nirmatrelvir (CYP3A4) — <i>CYP3A4: normal</i>
albendazole (CYP3A4; FMO) — <i>CYP3A4: normal</i>	erythromycin (CYP3A4) — <i>CYP3A4: normal</i>	nystatin (not absorbed)
amoxicillin (renal)	ethambutol (renal)	oseltamivir (CES1)
artemether + lumefantrine (CYP3A4) — <i>CYP3A4: normal</i>	famcyclovir (aldehyde oxidase)	piperacillin + tazobactam (renal)
azithromycin (biliary)	phenoxymethylpenicillin (renal)	pyrazinamide (amidease; xanthine oxidase)
cephalexin (renal)	fluconazole (renal; inhibits CYP2C9/2C19/3A4) <i>CYP2C9: IM (*1/*3)</i>	polymyxin B (topic/renal)
cefazolin (renal)	phosphomycin (renal)	posaconazole (UGT1A4)
cefdinir (renal)	isavuconazole (CYP3A4) — <i>CYP3A4: normal</i>	rifampicin (esterases; inducer)
cefixime (renal)	itraconazole (CYP3A4) <i>CYP3A4: normal</i>	rilpivirine (CYP3A4) — <i>CYP3A4: normal</i>
cefuroxime (renal)	ivermectin (CYP3A4) — <i>CYP3A4: normal</i>	ritonavir (CYP3A4) <i>CYP3A4: normal</i>
ketoconazole (CYP3A4) — <i>CYP3A4: normal</i>	© 2011 Elsevier. (renal)	tenofovir (renal)
clarithromycin (CYP3A4 (inhibitor)) <i>CYP3A4: normal</i>	linezolid (non-CYP oxidation)	trimethoprim (renal)
clindamycin (CYP3A4) — <i>CYP3A4: normal</i>	mebendazole (hepatic)	valacyclovir (renal)
clotrimazole (topic)	mefloquine (CYP3A4) — <i>CYP3A4: normal</i>	valganciclovir (renal)
darunavir (CYP3A4) — <i>CYP3A4: normal</i>	meropenem (renal)	vancomycin (renal)
doxycycline (no relevant metabolism)	metronidazole (CYP2A6; hepatic) <i>CYP2A6: probable normal</i>	
emtricitabine (renal)	molnupiravir (hydrolysis)	

3.7 Oncology — 39 medicines

Monitoring: 6 · indeterminate: 1 · normal: 13 · without known pharmacogenetic interaction: 19

Medication	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
belinostat Monitoring	UGT1A1 : IM (*1/*28)	exposure something bigger	FDA label: *28/*28 → reduced initial dose; *1/*28 without adjustment, monitoring toxicity.	Label FDA; CPIC (B) <i>moderate</i>	UGT1A1: high
irinotecan Monitoring	UGT1A1 : IM (*1/*28) DPYD : normal (AS 2)	something greater risk of neutropenia	FDA label: *1/*28 (IM) has increased risk of neutropenia; dose reduction recommended only for *28/*28. DPWG: *1/*28 without action. To monitor the blood count closely.	Label FDA; DPWG; CPIC (in preparation) <i>strong (PharmGKB 1A)</i>	UGT1A1: high DPYD: high for the tested variants
nilotinib Monitoring	UGT1A1 : IM (*1/*28) HLA-B*57:01 : negative	hyperbilirubinemia more likely	FDA labels: UGT1A1*28 increases hyperbilirubinemia (benign, but confuses liver surveillance). For pazopanib, HLA-B*57:01 negative (less risk of high ALT).	FDA labels <i>moderate</i>	UGT1A1: high HLA-B*57:01
pazopanib Monitoring	UGT1A1 : IM (*1/*28) HLA-B*57:01 : negative	hyperbilirubinemia more likely	FDA labels: UGT1A1*28 increases hyperbilirubinemia (benign, but confuses liver surveillance). For pazopanib, HLA-B*57:01 negative (less risk of high ALT).	FDA labels <i>moderate</i>	UGT1A1: high HLA-B*57:01
satituzumab govitecan Monitoring	UGT1A1 : IM (*1/*28)	something greater risk of neutropenia	FDA label: *28/*28 with more neutropenia; heterozygotes with intermediate risk. Monitoring blood count.	FDA label <i>moderate</i>	UGT1A1: high
tamoxifen Monitoring	CYP2D6 : Probable NM (without CNV) F5/F2 : non carrier	probable normal - confirm CYP2D6 with CNV	CPIC 2018 NM → tamoxifen 20 mg/day and avoid moderate/strong inhibitors of CYP2D6 (paroxetine, fluoxetine, bupropion). If an *5 deletion turned it into IM, the recommendation would change (major dose or aromatase inhibitor) → confirm with CNV before. F5/F2 normal.	CPIC 2018; DPWG; FDA label <i>strong (CPIC A; PharmGKB 1A)</i>	CYP2D6: moderate F5/F2: high
lapatinib Undetermined	HLA-DRB1*07:01 : indeterminate	indeterminate	FDA label: HLA-DRB1*07:01/DQA1*02:01 are associated with hepatotoxicity; not evaluated in this chip.	FDA label <i>moderate</i>	HLA-DRB1*07:01: non-determinable
capecitabine Normal	DPYD : normal (AS 2)	normal	CPIC 2023/DPWG: DPYD AS 2 → label dose. The 13 variants tested include the reference 4 (CPIC/EMA). Before chemotherapy, EMA recommends clinical DPD test; repeat because rare variants are not seen in the chip. (It also applies to the topic 5-FU.)	CPIC 2023; DPWG; EMA 2020; FDA label <i>strong (CPIC A; PharmGKB 1A)</i>	DPYD: high for the tested variants
cyclophosphamide Normal	CYP2B6 : RM	normal	CYP2B6 participates in metabolism/activation, without clinical recommendation by genotype. Usual dose.	PharmGKB (3 level); label FDA (prasugrel) <i>weak</i>	CYP2B6: moderate
cisplatin Normal	TPMT : NM	normal	The label FDA refers to variants TPMT associated with ototoxicity in children; without variants TPMT. Normal auditory surveillance.	FDA label; PharmGKB <i>weak to moderate</i>	TPMT: high
dabrafenib Normal	G6PD : normal	normal	CPIC 2022 / labels: Normal G6PD → with no reason to avoid hemolysis risk. Rare variants of G6PD not excluded.	CPIC 2022 (G6PD); FDA labels <i>strong (CPIC A)</i>	G6PD: high for the tested variants
doxorubicin Normal	G6PD : normal	normal	CPIC 2022 / labels: Normal G6PD → with no reason to avoid hemolysis risk. Rare variants of G6PD not excluded.	CPIC 2022 (G6PD); FDA labels <i>strong (CPIC A)</i>	G6PD: high for the tested variants
erdafitinib Normal	CYP2C9 : IM (*1/*3)	normal	CYP2C9 contributes to metabolism, but label recommendations apply only to slow metabolizers (*3/*3) or associations with inhibitors. Usual dose.	FDA/EMA labels <i>moderate</i>	CYP2C9: high

Medication	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
fluorouracil (5-FU) Normal	DPYD : normal (AS 2)	normal	CPIC 2023/DPWG: DPYD AS 2 → label dose. The 13 variants tested include the reference 4 (CPIC/EMA). Before chemotherapy, EMA recommends clinical DPD test; repeat because rare variants are not seen in the chip. (It also applies to the topic 5-FU.)	CPIC 2023; DPWG; EMA 2020; FDA label <i>strong</i> (CPIC A; PharmGKB 1A)	DPYD: high for the tested variants
gefitinib Normal	CYP2D6 : Probable NM (without CNV)	normal	The CYP2D6 is on the label/guidelines, but the recommendations apply to slow or ultrafast metabolizers. Probable NM → usual dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	FDA/EMA labels; CPIC 2021 (hydrocodone/oxycodone: no action for NM) <i>moderate</i>	CYP2D6: moderate
mercaptopurin Normal	TPMT : NM NUDT15 : NM	normal	CPIC 2018 TPMT NM + NUDT15 NM → standard initial dose; adjust by myelosuppression as usual.	CPIC 2018; DPWG; labels FDA <i>strong</i> (CPIC A; PharmGKB 1A)	TPMT: high NUDT15: high
rasburicase Normal	G6PD : normal	normal	CPIC 2022 / labels: Normal G6PD → with no reason to avoid hemolysis risk. Rare variants of G6PD not excluded.	CPIC 2022 (G6PD); FDA labels <i>strong</i> (CPIC A)	G6PD: high for the tested variants
rucaparib Normal	CYP2D6 : Probable NM (without CNV)	normal	The CYP2D6 is on the label/guidelines, but the recommendations apply to slow or ultrafast metabolizers. Probable NM → usual dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	FDA/EMA labels; CPIC 2021 (hydrocodone/oxycodone: no action for NM) <i>moderate</i>	CYP2D6: moderate
tegafur (with gimeracil/oteracyl) Normal	DPYD : normal (AS 2)	normal	CPIC 2023/DPWG: DPYD AS 2 → label dose. The 13 variants tested include the reference 4 (CPIC/EMA). Before chemotherapy, EMA recommends clinical DPD test; repeat because rare variants are not seen in the chip. (It also applies to the topic 5-FU.)	CPIC 2023; DPWG; EMA 2020; FDA label <i>strong</i> (CPIC A; PharmGKB 1A)	DPYD: high for the tested variants
thioguanine Normal	TPMT : NM NUDT15 : NM	normal	CPIC 2018 TPMT NM + NUDT15 NM → standard initial dose; adjust by myelosuppression as usual.	CPIC 2018; DPWG; labels FDA <i>strong</i> (CPIC A; PharmGKB 1A)	TPMT: high NUDT15: high

No known pharmacogenetic interaction (19):

abemaciclib (CYP3A4) — CYP3A4: *normal*
 abiraterone (CYP3A4; SULT2A1) — CYP3A4: *normal*
 anastrozole (UGT1A4; CYP3A4) — CYP3A4: *normal*
 bicalutamide (CYP3A4) — CYP3A4: *normal*
 carboplatin (renal)
 docetaxel (CYP3A4) — CYP3A4: *normal*
 enzalutamide (CYP2C8; CYP3A4) — CYP3A4: *normal*; CYP2C8: *normal*

erlotinib (CYP3A4; CYP1A2) — CYP3A4: *normal*; CYP1A2: *1F *het.*
 exemestane (CYP3A4) — CYP3A4: *normal*
 imatinib (CYP3A4) — CYP3A4: *normal*
 letrozole (CYP2A6; CYP3A4) — CYP3A4: *normal*; CYP2A6: *probable normal*
 nivolumab (proteolysis)
 olaparib (CYP3A4) — CYP3A4: *normal*

osimertinib (CYP3A4) — CYP3A4: *normal*
 oxaliplatin (non-enzymatic)
 paclitaxel (CYP2C8; CYP3A4) — CYP3A4: *normal*; CYP2C8: *normal*
 palbociclib (CYP3A4; SULT2A1) — CYP3A4: *normal*
 pembrolizumab (proteolysis)
 trastuzumab (proteolysis)

3.8 Anesthesia and perioperative — 25 medications

normal: 18 · without known pharmacogenetic interaction: 7

Medication	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
articain Normal	G6PD : normal CYB5R : indeterminate	normal	Labels: risk of G6PD/CYB5R deficit methemoglobinemia. Normal G6PD; CYB5R non-testable (rare). Usual dose.	FDA labels <i>moderate</i>	G6PD: high for the tested variants CYB5R: non-determinable
bupivacaine Normal	G6PD : normal CYB5R : indeterminate	normal	Labels: risk of G6PD/CYB5R deficit methemoglobinemia. Normal G6PD; CYB5R non-testable (rare). Usual dose.	FDA labels <i>moderate</i>	G6PD: high for the tested variants CYB5R: non-determinable
ketamine Normal	CYP2B6 : RM	normal	CYP2B6 participates in metabolism/activation, without clinical recommendation by genotype. Usual dose.	PharmGKB (3 level); label FDA (prasugrel) <i>weak</i>	CYP2B6: moderate
desflurane Normal	RYR1/CACNA1S : no variants	normal (reduced risk, not absent)	CPIC 2018: no pathogenic variant among the tested 36 → standard precautions. A negative result does not exclude malignant hyperthermia; personal/family history continues to tell.	CPIC 2018; FDA labels <i>strong (CPIC A; PharmGKB 1A)</i>	RYR1/CACNA1S: partial
isoflurane Normal	RYR1/CACNA1S : no variants	normal (reduced risk, not absent)	CPIC 2018: no pathogenic variant among the tested 36 → standard precautions. A negative result does not exclude malignant hyperthermia; personal/family history continues to tell.	CPIC 2018; FDA labels <i>strong (CPIC A; PharmGKB 1A)</i>	RYR1/CACNA1S: partial
lidocaine Normal	G6PD : normal CYB5R : indeterminate	normal	Labels: risk of G6PD/CYB5R deficit methemoglobinemia. Normal G6PD; CYB5R non-testable (rare). Usual dose.	FDA labels <i>moderate</i>	G6PD: high for the tested variants CYB5R: non-determinable
midazolam Normal	CYP3A5 : expressor (*1/*3) CYP3A4 : normal	normal	CYP3A substrates; the effect of CYP3A5 expressor is small/unconsistent for these drugs and there is no guideline. Usual dose (cyclosporin/everolimus guided by levels, as always).	PharmGKB (3 level) <i>weak</i>	CYP3A5: high CYP3A4: high
mivacurium Normal	BCHE : normal	normal	FDA label: BCHE deficit prolongs the blockade. Normal BCHE.	FDA label <i>moderate</i>	BCHE: high
palonosetron Normal	CYP2D6 : Probable NM (without CNV)	normal	The CYP2D6 is on the label/guidelines, but the recommendations apply to slow or ultrafast metabolizers. Probable NM → usual dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2-5%), ultrafast.	FDA/EMA labels; CPIC 2021 (hydrocodone/oxycodone: no action for NM) <i>moderate</i>	CYP2D6: moderate
prilocaine Normal	G6PD : normal CYB5R : indeterminate	normal	Labels: risk of G6PD/CYB5R deficit methemoglobinemia. Normal G6PD; CYB5R non-testable (rare). Usual dose.	FDA labels <i>moderate</i>	G6PD: high for the tested variants CYB5R: non-determinable
propofol Normal	CYP2B6 : RM	normal	CYP2B6 participates in metabolism/activation, without clinical recommendation by genotype. Usual dose.	PharmGKB (3 level); label FDA (prasugrel) <i>weak</i>	CYP2B6: moderate
nitrogen protoxide Normal	MTHFR : 677CT/1298AC	normal (minor effect)	Protoxide inactive to vitamin B12; MTHFR 677 heterozygous increases the rise of homocysteine (weak evidence; severe risk only in severe MTHFR deficiency).	PharmGKB (3 level) <i>weak</i>	MTHFR: high

Medication	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
remifentanyl Normal	OPRM1 : A/G COMT : Val/Val	normal	CPIC 2021 (opioides): no OPRM1 or COMT recommendation. The answer.	CPIC 2021 <i>strong (non-action)</i>	OPRM1: high COMT: high
ropivacaine Normal	G6PD : normal CYB5R : indeterminate	normal	Labels: risk of G6PD/CYB5R deficit methemoglobinemia. Normal G6PD; CYB5R non-testable (rare). Usual dose.	FDA labels <i>moderate</i>	G6PD: high for the tested variants CYB5R: non-determinable
sevoflurane Normal	RYR1/CACNA1S : no variants	normal (reduced risk, not absent)	CPIC 2018: no pathogenic variant among the tested 36 → standard precautions. A negative result does not exclude malignant hyperthermia; personal/family history continues to tell.	CPIC 2018; FDA labels <i>strong (CPIC A; PharmGKB 1A)</i>	RYR1/CACNA1S: partial
sufentanil Normal	OPRM1 : A/G COMT : Val/Val	normal	CPIC 2021 (opioides): no OPRM1 or COMT recommendation. The answer.	CPIC 2021 <i>strong (non-action)</i>	OPRM1: high COMT: high
suxametonium (succinylcholine) Normal	RYR1/CACNA1S : no variants BCHE : normal	normal	CPIC 2018: no variant RYR1/CACNA1S → standard precautions. Normal BCHE → normal duration of blockade (without prolonged apnea).	CPIC 2018; label FDA (BCHE) <i>strong</i>	RYR1/CACNA1S: partial BCHE: high
tropisetrom Normal	CYP2D6 : Probable NM (without CNV) SLC22A1 (OCT1) : normal	normal	CPIC 2017 NM → standard dose. (Ultrarpids would have less efficacy.) Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	CPIC 2017; DPWG <i>strong (CPIC A; PharmGKB 1A)</i>	CYP2D6: moderate SLC22A1 (OCT1): high

No known pharmacogenetic interaction (7):

dexmedetomidine (UGT; CYP2A6) — *CYP2A6: probable normal*
droperidol (hepatic)
etomidate (sterases)

granisetron (CYP3A4) — *CYP3A4: normal*
neostigmine (cholinesterases)
rocuronium (biliary)

sugammadex (renal)

3.9 Endocrine and diabetes — 39 medicines

Monitoring: 6 · normal: 12 · without known pharmacogenetic interaction: 21

Medication	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
glibenclamide Monitoring	CYP2C9 : IM (*1/*3) G6PD : normal	higher exposure → more hypoglycemia	CYP2C9*3 increases sulfonylurea exposure (PharmGKB 2A-3); DPWG without mandatory action. Start low and monitor blood glucose. Normal G6PD: no risk of hemolysis.	PharmGKB; DPWG (no action); labels (G6PD) <i>weak to moderate</i>	CYP2C9: high G6PD: high for the tested variants
gliclazide Monitoring	CYP2C9 : IM (*1/*3) G6PD : normal	higher exposure → more hypoglycemia	CYP2C9*3 increases sulfonylurea exposure (PharmGKB 2A-3); DPWG without mandatory action. Start low and monitor blood glucose. Normal G6PD: no risk of hemolysis.	PharmGKB; DPWG (no action); labels (G6PD) <i>weak to moderate</i>	CYP2C9: high G6PD: high for the tested variants
glimepiride Monitoring	CYP2C9 : IM (*1/*3) G6PD : normal	higher exposure → more hypoglycemia	CYP2C9*3 increases sulfonylurea exposure (PharmGKB 2A-3); DPWG without mandatory action. Start low and monitor blood glucose. Normal G6PD: no risk of hemolysis.	PharmGKB; DPWG (no action); labels (G6PD) <i>weak to moderate</i>	CYP2C9: high G6PD: high for the tested variants
glipizide Monitoring	CYP2C9 : IM (*1/*3) G6PD : normal	higher exposure → more hypoglycemia	CYP2C9*3 increases sulfonylurea exposure (PharmGKB 2A-3); DPWG without mandatory action. Start low and monitor blood glucose. Normal G6PD: no risk of hemolysis.	PharmGKB; DPWG (no action); labels (G6PD) <i>weak to moderate</i>	CYP2C9: high G6PD: high for the tested variants
somatropin (growth hormone) Monitoring	TERT/melanoma (context)	precaution (oncologic genetic context)	It is not pharmacogenetic interaction: the genetic plan suggests avoiding GH/IGF-1 by risk TERT alleles and nevi in the 88th percentile (melanoma by the chip is p6, 5/p25) (suggestion, not applied; weak evidence).	Genetic plan (TERT, nevi) <i>weak</i>	—
tolbutamide Monitoring	CYP2C9 : IM (*1/*3) G6PD : normal	higher exposure → more hypoglycemia	CYP2C9*3 increases sulfonylurea exposure (PharmGKB 2A-3); DPWG without mandatory action. Start low and monitor blood glucose. Normal G6PD: no risk of hemolysis.	PharmGKB; DPWG (no action); labels (G6PD) <i>weak to moderate</i>	CYP2C9: high G6PD: high for the tested variants
★ acarbose Normal	—	normal	Without known PGx (acts in the intestine, it is almost not absorbed).	— —	—
★ dapagliflozin Normal	UGT1A9 : normal	normal	Main normal UGT1A9 (without *3) → normal exposure.	PharmGKB <i>weak</i>	UGT1A9: high
dulaglutide Normal	GLP1R (informative)	normal	Peptides without CYP metabolism; without actionable PGx. (GLP1R variants associated with response only in exploratory studies.)	PharmGKB (3 level) <i>weak</i>	—
exenatide Normal	GLP1R (informative)	normal	Peptides without CYP metabolism; without actionable PGx. (GLP1R variants associated with response only in exploratory studies.)	PharmGKB (3 level) <i>weak</i>	—
★ imeglumine Normal	SLC22A2 (OCT2) : A270S het. SLC22A1 (OCT1) : normal	normal	Excreted by the kidney via OCT2/MATE; OCT2 A270S heterozygous, effect not studied and probably irrelevant. Normal OCT1.	Genetic plan <i>weak</i>	SLC22A2 (OCT2): high SLC22A1 (OCT1): high
liraglutide Normal	GLP1R (informative)	normal	Peptides without CYP metabolism; without actionable PGx. (GLP1R variants associated with response only in exploratory studies.)	PharmGKB (3 level) <i>weak</i>	—
Metformin Normal	SLC22A1 (OCT1) : normal	normal	OCT1 of normal function (without the 4 variants of loss of function) → normal uptake/depuration. No guideline.	PharmGKB (2B-3 level) <i>weak</i>	SLC22A1 (OCT1): high
nateglinide Normal	CYP2C9 : IM (*1/*3)	normal	FDA label: only slow metabolizers (*3/*3) have clinically higher exposure. With *1/*3, usual dose.	FDA label <i>moderate</i>	CYP2C9: high

Medication	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
repaglinide Normal	SLCO1B1 : increased function CYP2C8 : normal	normal	SLCO1B1 (augmented function) and normal CYP2C8; the labels refer only to 521CC. Usual dose.	FDA labels; PharmGKB <i>weak</i>	SLCO1B1: high CYP2C8: high
★ retatrutide Normal	GLP1R (informative)	normal	Peptides without CYP metabolism; without actionable PGx. (GLP1R variants associated with response only in exploratory studies.)	PharmGKB (3 level) <i>weak</i>	—
semaglutide Normal	GLP1R (informative)	normal	Peptides without CYP metabolism; without actionable PGx. (GLP1R variants associated with response only in exploratory studies.)	PharmGKB (3 level) <i>weak</i>	—
tirzepatido Normal	GLP1R (informative)	normal	Peptides without CYP metabolism; without actionable PGx. (GLP1R variants associated with response only in exploratory studies.)	PharmGKB (3 level) <i>weak</i>	—

No known pharmacogenetic interaction (21):

cabergoline (hydrolysis)	insulin detection (proteolysis)	propylthiouracil (UGT)
cinacalcete (CYP3A4; CYP1A2) — <i>CYP3A4: normal; CYP1A2: *1F het.</i>	insulin glargine (proteolysis)	sitagliptin (renal)
denosumab (proteolysis)	human insulin (regular / isophane NPH) (proteolysis)	teriparatide (proteolysis)
desmopressin (renal)	insulin lispro (proteolysis)	thiamazole (matimazol) (hepatic)
empagliflozin (UGT2B7/1A3/1A9)	linagliptin (biliary)	vildagliptin (hydrolysis)
phentermine (renal)	pioglitazone (CYP2C8; CYP3A4) — <i>CYP3A4: normal; CYP2C8: normal</i>	rosedronic acid (renal)
insulin as part (proteolysis)	not recommended for other genetic reasons (p94 bladder cancer, 9 + slow NAT2; ejection fraction p92)	zoledronic acid (renal)
insulin degludec (proteolysis)		

3.10 Gastroenterology — 35 medicines

Monitoring: 1 · normal: 10 · without known pharmacogenetic interaction: 24

Medication	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
sulfasalazine Monitoring	NAT2: slow G6PD: normal	more adverse effects	Slow acetylators accumulate sulfapyridine: more nausea, headache, hemolysis and skin reactions (FDA label). Starting low and full-time. Normal G6PD.	FDA label; PharmGKB moderate	NAT2: high G6PD: high for the tested variants
dexlansoprazole Normal	CYP2C19: NM	normal	CPIC 2020: normal metabolizer → standard initial dose; for H. pylori or erosive esophagitis consider +50–100% and monitor efficacy (it is the general recommendation for NM).	CPIC 2020; DPWG moderate (CPIC A; PharmGKB 1A)	CYP2C19: high
esomeprazole Normal	CYP2C19: NM	normal	CPIC 2020: normal metabolizer → standard initial dose; for H. pylori or erosive esophagitis consider +50–100% and monitor efficacy (it is the general recommendation for NM).	CPIC 2020; DPWG moderate (CPIC A; PharmGKB 1A)	CYP2C19: high
lansoprazole Normal	CYP2C19: NM	normal	CPIC 2020: normal metabolizer → standard initial dose; for H. pylori or erosive esophagitis consider +50–100% and monitor efficacy (it is the general recommendation for NM).	CPIC 2020; DPWG moderate (CPIC A; PharmGKB 1A)	CYP2C19: high
meclizine Normal	CYP2D6: Probable NM (without CNV)	normal	The CYP2D6 is on the label/guidelines, but the recommendations apply to slow or ultrafast metabolizers. Probable NM → usual dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	FDA/EMA labels; CPIC 2021 (hydrocodone/oxycodone: no action for NM) moderate	CYP2D6: moderate
metoclopramide Normal	CYP2D6: Probable NM (without CNV)	normal	The CYP2D6 is on the label/guidelines, but the recommendations apply to slow or ultrafast metabolizers. Probable NM → usual dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	FDA/EMA labels; CPIC 2021 (hydrocodone/oxycodone: no action for NM) moderate	CYP2D6: moderate
omeprazole Normal	CYP2C19: NM	normal	CPIC 2020: normal metabolizer → standard initial dose; for H. pylori or erosive esophagitis consider +50–100% and monitor efficacy (it is the general recommendation for NM).	CPIC 2020; DPWG moderate (CPIC A; PharmGKB 1A)	CYP2C19: high
ondansetron Normal	CYP2D6: Probable NM (without CNV) SLC22A1 (OCT1): normal	normal	CPIC 2017 NM → standard dose. (Ultrarpids would have less efficacy.) Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2–5%), ultrafast.	CPIC 2017; DPWG strong (CPIC A; PharmGKB 1A)	CYP2D6: moderate SLC22A1 (OCT1): high
pantoprazole Normal	CYP2C19: NM	normal	CPIC 2020: normal metabolizer → standard initial dose; for H. pylori or erosive esophagitis consider +50–100% and monitor efficacy (it is the general recommendation for NM).	CPIC 2020; DPWG moderate (CPIC A; PharmGKB 1A)	CYP2C19: high
rabeprazole Normal	CYP2C19: NM	normal	CPIC 2020: normal metabolizer → standard initial dose; for H. pylori or erosive esophagitis consider +50–100% and monitor efficacy (it is the general recommendation for NM).	CPIC 2020; DPWG moderate (CPIC A; PharmGKB 1A)	CYP2C19: high
vonoprazane Normal	CYP2C19: NM	normal	CYP2C19 is on the label; normal metabolizer → usual dose.	FDA label moderate	CYP2C19: high

No known pharmacogenetic interaction (24):

aprepitant (CYP3A4) — CYP3A4: normal
bisacodyl (hydrolysis)
butylscopolamine (hydrolysis)

dicyclomine (hepatic)
docusate (no relevant metabolism)
domperidone (CYP3A4) — CYP3A4: normal

famotidine (renal)
infliximab (proteolysis)
lactulose (intestinal flower)

linaclotide (not absorbed)

loperamide (CYP3A4; CYP2C8) — *CYP3A4: normal; CYP2C8: normal*

macrogol (polyethylene glycol) (not absorbed)

mebeverin (sterases)

mesalazine (NAT1)

pancrelipase (pancreatic enzymes) (not absorbed)

prucalopride (renal)

racecadotril (hydrolysis)

rifaximine (not absorbed)

sene (sene) (intestinal flower)

sulfasalazine (not absorbed)

trimebutin (hepatic)

ustecinumab (proteolysis)

vedolizumab (proteolysis)

ursodeoxycholic acid (biliary)

3.11 Respiratory and allergy — 42 medicines

indeterminate: 1 · normal: 6 · without known pharmacogenetic interaction: 35

Medication	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
ivacaftor (and combinations with elexacaftor/ tezacaftor) Undetermined	CFTR : not applicable	not applicable	Indicated only in cystic fibrosis with specific CFTR variants (CPIC 2014). It does not apply.	CPIC 2014; labels <i>strong (for FQ patients)</i>	CFTR: partial
dextromethorphan Normal	CYP2D6 : Probable NM (without CNV)	normal	The CYP2D6 is on the label/guidelines, but the recommendations apply to slow or ultrafast metabolizers. Probable NM → usual dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2-5%), ultrafast.	FDA/EMA labels; CPIC 2021 (hydrocodone/oxycodone: no action for NM) <i>moderate</i>	CYP2D6: moderate
formoterol Normal	CYP2D6 : Probable NM (without CNV)	normal	The CYP2D6 is on the label/guidelines, but the recommendations apply to slow or ultrafast metabolizers. Probable NM → usual dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2-5%), ultrafast.	FDA/EMA labels; CPIC 2021 (hydrocodone/oxycodone: no action for NM) <i>moderate</i>	CYP2D6: moderate
indacaterol Normal	UGT1A1 : IM (*1/*28)	normal	Glucuronidated by UGT1A1; the labels mention only *28/*28 (mild exposure increase, without adjustment). Usual dose.	FDA labels <i>moderate</i>	UGT1A1: high
promethazine Normal	CYP2D6 : Probable NM (without CNV)	normal	The CYP2D6 is on the label/guidelines, but the recommendations apply to slow or ultrafast metabolizers. Probable NM → usual dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2-5%), ultrafast.	FDA/EMA labels; CPIC 2021 (hydrocodone/oxycodone: no action for NM) <i>moderate</i>	CYP2D6: moderate
theophylline Normal	CYP1A2 : *1F het.	normal	CYP1A2*1F heterozygous: intermediate inductibility (the effect is noted mainly in smokers). No guideline; usual dose (clozapine and theophylline by levels, as always).	PharmGKB (3 level) <i>weak</i>	CYP1A2: high
umeclidinium Normal	CYP2D6 : Probable NM (without CNV)	normal	The CYP2D6 is on the label/guidelines, but the recommendations apply to slow or ultrafast metabolizers. Probable NM → usual dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2-5%), ultrafast.	FDA/EMA labels; CPIC 2021 (hydrocodone/oxycodone: no action for NM) <i>moderate</i>	CYP2D6: moderate

No known pharmacogenetic interaction (35):

acetylcysteine (deacetylation)	desloratadine (CYP2C8; UGT2B10) — <i>CYP2C8: normal</i>	loratadine (CYP3A4; CYP2D6) — <i>CYP3A4: normal; CYP2D6: Probable NM (without CNV)</i>
aclidinium (sterases)	diphenhydramine (CYP2D6) — <i>CYP2D6: Probable NM (without CNV)</i>	mepolizumab (proteolysis)
ambroxol (CYP3A4) — <i>CYP3A4: normal</i>	ebastin (CYP3A4; CYP2J2) — <i>CYP3A4: normal</i>	mometasone (CYP3A4) — <i>CYP3A4: normal</i>
azelastin (CYP3A4; CYP2D6) — <i>CYP3A4: normal; CYP2D6: Probable NM (without CNV)</i>	fexophenadine (OATP; P-gp)	montelukast (CYP2C8; CYP2C9) — <i>CYP2C9: IM (*1/*3); CYP2C8: normal</i>
beclomethasone (sterases; CYP3A4) <i>CYP3A4: normal</i>	fluticasone (CYP3A4) — <i>CYP3A4: normal</i>	olopatadine (renal)
benzonatate (sterases)	glucopyrronium (renal)	omalizumab (proteolysis)
bilastine (no metabolism)	guaifenesin (hepatic)	pseudoephedrine (renal)
bromopheniramine (hepatic)	hydroxyzine (CYP3A4) — <i>CYP3A4: normal</i>	roflumilast (CYP3A4; CYP1A2) — <i>CYP3A4: normal; CYP1A2: *1F het.</i>
budesonide (CYP3A4) — <i>CYP3A4: normal</i>	ipratropium (hydrolysis)	rupatadine (CYP3A4) — <i>CYP3A4: normal</i>
cetirizine (renal)	levocetirizine (renal)	salbutamol (SULT1A3)
ciproheptadine (UGT)	levosalbutamol (SULT1A3)	

salmeterol (CYP3A4) — *CYP3A4: normal*

thiotropium (renal)

vilanterol (CYP3A4) — *CYP3A4: normal*

3.12 Immunology, rheumatology and transplantation — 25 medicines

dose adjustment: 1 · Monitoring: 3 · normal: 8 · without known pharmacogenetic interaction: 13

Medication	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
tacrolimus (systemic; the topic is not relevant PGx) Adjust dose	CYP3A5 :expressor (*1/*3) CYP3A4 : normal	lower exposure → initial dose 1.5–2	CPIC 2015: CYP3A5 (*1/*3) → increase the initial dose 1.5–2× (without exceeding the maximum of the label) and guide by levels. Only relevant for systemic tacrolimus (transplantation, autoimmunity); the topic is not affected.	CPIC 2015; DPWG; FDA label <i>strong (CPIC A; PharmGKB 1A)</i>	CYP3A5: high CYP3A4: high
leflunomide Monitoring	PNPLA3/Pi*MZ (liver) : unfavourable	monitoring (liver)	No PGx guideline. Infavourable liver genetic context (PNPLA3, MBOAT7, probable Pi*MZ): is not a PGx guideline, but justifies surveillance of transaminases.	Indirect (hepatic genetic) <i>weak/indirect</i>	PNPLA3/Pi*MZ (liver): moderate
methotrexate Monitoring	MTHFR : 677CT/1298AC SLCO1B1 : increased function	toxicity possibly something greater (weak evidence)	MTHFR 677 C/T: weak and inconsistent association with more toxicity (CPIC level C; DPWG without action). Support folate in the form 5-MTHF (non-folic acid; MTHFR 677CT/1298AC) and usual surveillance (hemogram, liver). Infavourable liver genetic context (PNPLA3, MBOAT7, probable Pi*MZ): is not a PGx guideline, but justifies surveillance of transaminases.	CPIC (C); DPWG (no action) <i>weak</i>	MTHFR: high SLCO1B1: high
★ sirolimus (rapamycin) Monitoring	CYP3A5 :expressor (*1/*3) CYP3A4 : normal	exposure possibly lower per mg	No guideline for sirolimus. In transplantation, CYP3A5 expressers have higher clearance and lower levels per mg (moderate evidence). Normal CYP3A4. A sirolimus dosage to calibrate the dose (suggestion, not applied).	PharmGKB (2B–3 level); studies in transplantation <i>moderate to weak</i>	CYP3A5: high CYP3A4: high
allopurinol Normal	HLA-B*58:01 : probable negative ABCG2 : normal	probable normal - typing before is good practice	CPIC 2015 B*58:01 negative dose. The negativity comes from an imputed tag (calibrated in few carriers), so the clinical HLA-B*58:01 typing remains good practice. Normal ABCG2 → normal response.	CPIC 2015; DPWG; ACR 2020 <i>strong (CPIC A; PharmGKB 1A)</i>	HLA-B*58:01: moderate ABCG2: high
azathioprine Normal	TPMT : NM NUDT15 : NM	normal	CPIC 2018 TPMT NM + NUDT15 NM → standard initial dose; adjust by myelosuppression as usual.	CPIC 2018; DPWG; labels FDA <i>strong (CPIC A; PharmGKB 1A)</i>	TPMT: high NUDT15: high
ciclosporin Normal	CYP3A5 :expressor (*1/*3) CYP3A4 : normal	normal	CYP3A substrates; the effect of CYP3A5 expresser is small/unconsistent for these drugs and there is no guideline. Usual dose (cyclosporin/everolimus guided by levels, as always).	PharmGKB (3 level) <i>weak</i>	CYP3A5: high CYP3A4: high
everolimus Normal	CYP3A5 :expressor (*1/*3) CYP3A4 : normal	normal	CYP3A substrates; the effect of CYP3A5 expresser is small/unconsistent for these drugs and there is no guideline. Usual dose (cyclosporin/everolimus guided by levels, as always).	PharmGKB (3 level) <i>weak</i>	CYP3A5: high CYP3A4: high
hydroxychloroquine Normal	G6PD : normal	normal	CPIC 2022 / labels: Normal G6PD → with no reason to avoid hemolysis risk. Rare variants of G6PD not excluded.	CPIC 2022 (G6PD); FDA labels <i>strong (CPIC A)</i>	G6PD: high for the tested variants
pegloticosis Normal	G6PD : normal	normal	CPIC 2022 / labels: Normal G6PD → with no reason to avoid hemolysis risk. Rare variants of G6PD not excluded.	CPIC 2022 (G6PD); FDA labels <i>strong (CPIC A)</i>	G6PD: high for the tested variants
probenecid Normal	G6PD : normal	normal	CPIC 2022 / labels: Normal G6PD → with no reason to avoid hemolysis risk. Rare variants of G6PD not excluded.	CPIC 2022 (G6PD); FDA labels <i>strong (CPIC A)</i>	G6PD: high for the tested variants

Medication	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
upadacitinib Normal	CYP2D6 : Probable NM (without CNV)	normal	The CYP2D6 is on the label, without dose adjustment for normal metabolizers. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2-5%), ultrafast.	FDA labels <i>moderate</i>	CYP2D6: moderate

No known pharmacogenetic interaction (13):

adalimumab (proteolysis)	etanercept (proteolysis)	rituximab (proteolysis)
apremilaste (CYP3A4) — CYP3A4: normal	febuxostat (UGT; CYP1A2/2C8/2C9) — CYP1A2: *1F het.	secucinumab (proteolysis)
baricitinib (renal; CYP3A4) CYP3A4: normal	mycophenolate (mofethyl / mycophenolic acid) (UGT1A9; HPRT1 (via))	tocilizumab (proteolysis)
certolizumab pegole (proteolysis)	— UGT1A9: normal	tofacitinib (CYP3A4; CYP2C19) — CYP3A4: normal; CYP2C19: NM
colchicine (CYP3A4; P-gp) — CYP3A4: normal	risancizumab (proteolysis)	

3.13 Urology and sexual health — 21 medicines

normal: 13 · without known pharmacogenetic interaction: 8

Medication	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
avanafil Normal	CYP3A4 : normal	normal	Substrates of CYP3A4, which is normal (without *22). No actionable PGx.	PharmGKB <i>weak</i>	CYP3A4: high
dapoxetine Normal	CYP2D6 : Probable NM (without CNV)	normal	The CYP2D6 is on the label/guidelines, but the recommendations apply to slow or ultrafast metabolizers. Probable NM → usual dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2-5%), ultrafast.	FDA/EMA labels; CPIC 2021 (hydrocodone/oxycodone: no action for NM) <i>moderate</i>	CYP2D6: moderate
darifenacin Normal	CYP2D6 : Probable NM (without CNV)	normal	The CYP2D6 is on the label/guidelines, but the recommendations apply to slow or ultrafast metabolizers. Probable NM → usual dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2-5%), ultrafast.	FDA/EMA labels; CPIC 2021 (hydrocodone/oxycodone: no action for NM) <i>moderate</i>	CYP2D6: moderate
phenazopyridine Normal	G6PD : normal	normal	CPIC 2022 / labels: Normal G6PD → with no reason to avoid hemolysis risk. Rare variants of G6PD not excluded.	CPIC 2022 (G6PD); FDA labels <i>strong (CPIC A)</i>	G6PD: high for the tested variants
fesoterodine Normal	CYP2D6 : Probable NM (without CNV)	normal	The CYP2D6 is on the label/guidelines, but the recommendations apply to slow or ultrafast metabolizers. Probable NM → usual dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2-5%), ultrafast.	FDA/EMA labels; CPIC 2021 (hydrocodone/oxycodone: no action for NM) <i>moderate</i>	CYP2D6: moderate
flibanserin Normal	CYP2C19 : NM	normal	The CYP2C19 is a relevant route and is included in the label/guidelines, but the recommendations apply to intermediate/slow metabolizers. Normal metabolizer → usual dose.	Labels FDA/EMA; CPIC (level B/C) <i>moderate</i>	CYP2C19: high
mirabegron Normal	CYP2D6 : Probable NM (without CNV)	normal	The CYP2D6 is on the label/guidelines, but the recommendations apply to slow or ultrafast metabolizers. Probable NM → usual dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2-5%), ultrafast.	FDA/EMA labels; CPIC 2021 (hydrocodone/oxycodone: no action for NM) <i>moderate</i>	CYP2D6: moderate
sildenafil Normal	CYP3A4 : normal	normal	Substrates of CYP3A4, which is normal (without *22). No actionable PGx.	PharmGKB <i>weak</i>	CYP3A4: high
★ tadalafil Normal	CYP3A4 : normal	normal	Substrates of CYP3A4, which is normal (without *22). No actionable PGx.	PharmGKB <i>weak</i>	CYP3A4: high
tamsulosine Normal	CYP2D6 : Probable NM (without CNV)	normal	The CYP2D6 is on the label/guidelines, but the recommendations apply to slow or ultrafast metabolizers. Probable NM → usual dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2-5%), ultrafast.	FDA/EMA labels; CPIC 2021 (hydrocodone/oxycodone: no action for NM) <i>moderate</i>	CYP2D6: moderate
tolterodine Normal	CYP2D6 : Probable NM (without CNV)	normal	The CYP2D6 is on the label/guidelines, but the recommendations apply to slow or ultrafast metabolizers. Probable NM → usual dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2-5%), ultrafast.	FDA/EMA labels; CPIC 2021 (hydrocodone/oxycodone: no action for NM) <i>moderate</i>	CYP2D6: moderate
tropium Normal	CYP2D6 : Probable NM (without CNV)	normal	The CYP2D6 is on the label, without dose adjustment for normal metabolizers. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2-5%), ultrafast.	FDA labels <i>moderate</i>	CYP2D6: moderate

Medication	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
varденафил Normal	CYP3A4: normal	normal	Substrates of CYP3A4, which is normal (without *22). No actionable PGx.	PharmGKB weak	CYP3A4: high

No known pharmacogenetic interaction (8):

alfuzosin (CYP3A4) — CYP3A4: normal	finasteride (CYP3A4) — CYP3A4: normal	solifenacin (CYP3A4) — CYP3A4: normal
doxazosin (CYP3A4) — CYP3A4: normal	oxybutynin (CYP3A4) — CYP3A4: normal	terazosin (hepatic)
dutasteride (CYP3A4) — CYP3A4: normal	silodosin (UGT2B7; CYP3A4) — CYP3A4: normal	

3.14 Dermatology — 19 medicines

Monitoring: 2 · normal: 1 · without known pharmacogenetic interaction: 16

Medication	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
acitretin Monitoring	PNPLA3/Pi*MZ (liver): unfavourable	monitoring (liver)	No PGx guideline. Infavourable liver genetic context (PNPLA3, MBOAT7, probable Pi*MZ): is not a PGx guideline, but justifies surveillance of transaminases.	Indirect (hepatic genetic) weak/indirect	PNPLA3/Pi*MZ (liver): moderate
★ isotretinoin Monitoring	CYP2C8: normal CYP2C9: IM (*1/*3) CYP3A4: normal PNPLA3/Pi*MZ (liver): unfavourable	compatible - monitoring liver/lipids	No PGx guideline. Main route (CYP2C8) normal; CYP2C9*3 can slightly increase exposure. Infavourable liver genetic context (PNPLA3, MBOAT7, probable Pi*MZ): is not a PGx guideline, but justifies surveillance of transaminases. Daily photoprotection (p88 nevus; isotretinoin).	Genetic plan; PharmGKB weak	CYP2C8: high CYP2C9: high CYP3A4: high PNPLA3/Pi*MZ (liver): moderate
abrocitinib Normal	CYP2C19: NM	normal	The CYP2C19 is a relevant route and is included in the label/guidelines, but the recommendations apply to intermediate/slow metabolizers. Normal metabolizer → usual dose.	Labels FDA/EMA; CPIC (level B/C) moderate	CYP2C19: high

No known pharmacogenetic interaction (16):

adapalene (topic)	sulfur (topic) (topic)	benzoyl peroxide (topic)
calcipotriol (topic)	hydroquinone (topic)	sulfacetamide (topic)
cyclopyrox (topic)	imiquimod (topic)	tretinoin (CYP26; CYP2C8) — CYP2C8: normal
clobetasol (topic)	Ammonium lactate (topic) (topic)	azelaic acid (topic)
chlorhexidine (topic)	minocycline (hepatic)	
dupilumab (proteolysis)	mupirocin (topic)	

3.15 Hormones and corticosteroids — 31 medicines

normal: 9 · without known pharmacogenetic interaction: 22

Medication	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
ciproterone Normal	F5/F2: non carrier	normal (without common genetic thrombophilia)	DPWG / labels: F5 Leiden and prothrombin G20210A increase the risk of thrombosis with estrogens and others. Both are absent → without additional genetic risk for these variants.	DPWG (F5 Leiden–estrogens); FDA labels strong (for F5 Leiden)	F5/F2: high
drospirenone Normal	F5/F2: non carrier	normal (without common genetic thrombophilia)	DPWG / labels: F5 Leiden and prothrombin G20210A increase the risk of thrombosis with estrogens and others. Both are absent → without additional genetic risk for these variants.	DPWG (F5 Leiden–estrogens); FDA labels strong (for F5 Leiden)	F5/F2: high
estradiol Normal	F5/F2: non carrier	normal (without common genetic thrombophilia)	DPWG / labels: F5 Leiden and prothrombin G20210A increase the risk of thrombosis with estrogens and others. Both are absent → without additional genetic risk for these variants.	DPWG (F5 Leiden–estrogens); FDA labels strong (for F5 Leiden)	F5/F2: high
conjugated estrogens Normal	F5/F2: non carrier	normal (without common genetic thrombophilia)	DPWG / labels: F5 Leiden and prothrombin G20210A increase the risk of thrombosis with estrogens and others. Both are absent → without additional genetic risk for these variants.	DPWG (F5 Leiden–estrogens); FDA labels strong (for F5 Leiden)	F5/F2: high
ethinylestradiol Normal	F5/F2: non carrier	normal (without common genetic thrombophilia)	DPWG / labels: F5 Leiden and prothrombin G20210A increase the risk of thrombosis with estrogens and others. Both are absent → without additional genetic risk for these variants.	DPWG (F5 Leiden–estrogens); FDA labels strong (for F5 Leiden)	F5/F2: high
★ oxytocin Normal	OXTR (informative)	normal	No actionable PGx. OXTR rs53576 G/G is associated, inconsistently, with greater social sensitivity to oxytocin.	PharmGKB (3 level) weak	—
raloxifene Normal	UGT1A1: IM (*1/*28)	normal	Glucuronidated by UGT1A1; the labels mention only *28/*28 (mild exposure increase, without adjustment). Usual dose.	FDA labels moderate	UGT1A1: high
testosterone Normal	UGT2B17: indeterminate	normal	The deletion of UGT2B17 (not visible in the chip) changes the urinary excretion of testosterone (antidoping tests), not the clinical response.	PharmGKB weak	UGT2B17: non-determinable
tibolone Normal	F5/F2: non carrier	normal (without common genetic thrombophilia)	DPWG / labels: F5 Leiden and prothrombin G20210A increase the risk of thrombosis with estrogens and others. Both are absent → without additional genetic risk for these variants.	DPWG (F5 Leiden–estrogens); FDA labels strong (for F5 Leiden)	F5/F2: high

No known pharmacogenetic interaction (22):

betamethasone (CYP3A4) — CYP3A4: normal	levothyroxine (deiodation)	prednisolone (CYP3A4) — CYP3A4: normal
deogestrel (CYP3A4; CYP2C9) — CYP3A4: normal; CYP2C9: IM (*1/*3)	lyothyronine (deiodation)	prednisone (11β-HSD; CYP3A4) — CYP3A4: normal
dexamethasone (CYP3A4) — CYP3A4: normal	medroxyprogesterone (CYP3A4) — CYP3A4: normal	progesterone (CYP3A4; CYP2C19) — CYP3A4: normal; CYP2C19: NM
dienogest (CYP3A4) — CYP3A4: normal	methyprednisolone (CYP3A4) — CYP3A4: normal	thyroid desiccated (deiodation)
etonogestrel (CYP3A4) — CYP3A4: normal	norlegestromine (CYP3A4) — CYP3A4: normal	triamcinolone (CYP3A4) — CYP3A4: normal
fludrocortisone (hepatic)	norethisterone (CYP3A4) — CYP3A4: normal	ulipristal (CYP3A4) — CYP3A4: normal
hydrocortisone (CYP3A4) — CYP3A4: normal	norgestimate (CYP3A4) — CYP3A4: normal	
levonorgestrel (CYP3A4) — CYP3A4: normal	norgestrel (CYP3A4) — CYP3A4: normal	

3.16 Other (vitamins, minerals, ophthalmology, hematology, dependencies) — 31 medicines

Monitoring: 1 · normal: 8 · without known pharmacogenetic interaction: 22

Medication	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
eliglustat Monitoring	CYP2D6: Probable NM (without CNV)	Mandatory clinical genotyping	FDA/EMA label: the CYP2D6 phenotype defines the dose (NM: 84 mg 2×/ day). The chip does not exclude CNV → needs clinical testing.	Label FDA/EMA; CPIC (A/B) strong (compulsory test)	CYP2D6: moderate
avatrombopag Normal	CYP2C9: IM (*1/*3)	normal	CYP2C9 contributes to metabolism, but label recommendations apply only to slow metabolizers (*3/*3) or associations with inhibitors. Usual dose.	FDA/EMA labels moderate	CYP2C9: high
methylene blue Normal	G6PD: normal	normal	CPIC 2022 / labels: Normal G6PD → with no reason to avoid hemolysis risk. Rare variants of G6PD not excluded.	CPIC 2022 (G6PD); FDA labels strong (CPIC A)	G6PD: high for the tested variants
caffeine Normal	CYP1A2: *1F het.	normal	CYP1A2*1F heterozygous: intermediate inductibility (the effect is noted mainly in smokers). No guideline; usual dose (clozapine and theophylline by levels, as always).	PharmGKB (3 level) weak	CYP1A2: high
cevimelin Normal	CYP2D6: Probable NM (without CNV)	normal	The CYP2D6 is on the label/guidelines, but the recommendations apply to slow or ultrafast metabolizers. Probable NM → usual dose. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2-5%), ultrafast.	FDA/EMA labels; CPIC 2021 (hydrocodone/oxycodone: no action for NM) moderate	CYP2D6: moderate
eltrombopag Normal	F5/F2: non carrier	normal (without common genetic thrombophilia)	DPWG / labels: F5 Leiden and prothrombin G20210A increase the risk of thrombosis with estrogens and others. Both are absent → without additional genetic risk for these variants.	DPWG (F5 Leiden–estrogens); FDA labels strong (for F5 Leiden)	F5/F2: high
phytomenadione (vitamin K1) Normal	G6PD: normal	normal	CPIC 2022 / labels: Normal G6PD → with no reason to avoid hemolysis risk. Rare variants of G6PD not excluded.	CPIC 2022 (G6PD); FDA labels strong (CPIC A)	G6PD: high for the tested variants
timolol Normal	CYP2D6: Probable NM (without CNV)	normal	CPIC 2024 (betablockers): metoprolol in NM → standard dose; carvedilol/ nebivolol/propranolol without action for NM. FDA labels refer only to slow metabolizers. Without CNV evaluation: an *5 deletion (≈3% in the reference population) would make it an intermediate metabolizer; a duplication (≈2-5%), ultrafast.	CPIC 2024; DPWG; labels FDA moderate to strong (PharmGKB 1A for metoprolol)	CYP2D6: moderate
ascorbic acid (vitamin C) Normal	G6PD: normal	normal	CPIC 2022 / labels: Normal G6PD → with no reason to avoid hemolysis risk. Rare variants of G6PD not excluded.	CPIC 2022 (G6PD); FDA labels strong (CPIC A)	G6PD: high for the tested variants

No known pharmacogenetic interaction (22):

acetazolamide (renal)	calcium (sais) (no relevant metabolism)	sodium salts (bicarbonate, etc.) (no relevant metabolism)
epinephrine (COMT; MAO)	dorzolamide (topic)	ferrous sulfate (no relevant metabolism)
bimatoprost (topic)	tamarin (proteolysis)	thiamine (vitamin B1) (no relevant metabolism)
brimonidine (hepatic)	ergocalciferol (vitamin D2) (CYP2R1; CYP27B1)	botulinum toxin (proteolysis)
calcitriol (CYP24A1)	sodium fluoride (no relevant metabolism)	alendronic acid (renal)
cyanocobalamine (vitamin B12) (no relevant metabolism)	calcium phosphate (no relevant metabolism)	folic acid (DHFR; MTHFR) — MTHFR: 677CT/1298AC; use 5-MTHF, non-folic acid
potassium chloride (no relevant metabolism)	cannoprost (sterases)	
colecalfiferol (vitamin D3) (CYP2R1; CYP27B1)	magnesium salts (no relevant metabolism)	

4. Limitations

- **Chip + imputation, no sequencing.** Only the variants that the chip measures or that the imputation reconstructs with $DR2 \geq 0.8$ are seen. Rare ($<0.1\%$) and new variants are not seen. A 'normal' result means 'without the tested variants', and not 'without variants'.
- **CYP2D6:** deletions (*5), duplications and hybrid CYP2D6/2D7 are invisible. *3 and *6 are inferred and *17 is not attributable. About 5–8% of people with this pattern of SNPs would have another phenotype by CNV. For codeine, tramadol, tamoxifen, eligostat, pimozide and tetrabenazine, confirm by clinical test.
- **HLA:** B*58:01, A*31:01 and B*57:01 come from tags (chip/imputation) calibrated in 378 genomes of the reference population; B*58:01 is based on few carriers. B*15:02, A*33:01, DRB1*07:01 and rare alleles are not evaluated. When the label requires typing (abacavir) or it is good practice (allopurinol, carbamazepine), clinical typing is necessary.
- **Other gaps:** mtDNA (MT-RNR1) not determinable; CNVs of CYP2A6, UGT2B17, GSTM1/GSTT1; rare variants of G6PD, DPYD, RYR1, BCHE and CYB5R; repetitions.
- **Haplotype phase:** CYP2B6 *2/*4 and NAT2 *5/*5 assume the most probable phase.
- **Scope:** only genetics (without analysis or exams, on request). The interactions between drugs, renal/hepatic function and age change the real response and are not here. Phenoconversion: an inhibitor (e.g. fluoxetine, paroxetine or bupropion to CYP2D6; fluconazole to CYP2C9) can make a normal metabolizer slow.
- **List of medicines:** ClinCalc reflects the prescription in EUA (MEPS 2024); additions were chosen by clinical criteria, not from an official consumption list.
- **Evidence:** "no known pharmacogenetic interaction" means that there is no CPIC/DPWG guideline or germinal biomarker on the FDA label at the date (02-10-2026). It does not mean that genetics do not influence at all.
- **It is not a diagnosis.** It is a decision support to discuss with the doctor. The confirmation should be made with a clinical pharmacogenetic panel (preferably CYP2D6 with high resolution CNV and HLA). All indications are suggestion (not applied).

5. Sources and methods

- ClinCalc DrugStats Top 300 Drugs of 2024 (MEPS, AHRQ), discharged to 02-10-2026.
- CPIC — official API (gene-drug pairs and 2115 recommendations; automatic correspondence with the Sérgio F phenotype: 108 applicable recommendations). Guidelines: 2022 statins, NSAIDs 2020, warfarin 2017, phenytoin 2020, PPI 2020, SSRIs 2023, tricyclic 2016, opioids 2021, ondansetron 2017, betablockers 2024, atomoxetine 2019, tamoxifen 2018, clopidogrel 2022, tacrolimus 2015, fluoropyrimidine 2023, thiopurins 2018, G6PD 2022, HLA (abacavir 2014, allopurinol 2015, carbamazepine/oxcarbazepine 2017), anesthetic 2018, aminoglycosides 2021, efavirenz 2019, atazanavir 2015, voriconazole 2016, hydralazine (NAT2).
- DPWG (KNMP, Netherlands) — pharmacogenetic recommendations (coumarins, antidepressants, antipsychotics, PPI, statins, F5 Leiden–estrogens, among others).
- FDA — Table of Pharmacogenomic Biomarkers in Drug Labeling and Table of Pharmacogenetic Associations (discharged to 02-10-2026); labels FDA/EMA of the cited drugs.
- ClinPGx/PharmGKB — levels of evidence of clinical notes (the site was in maintenance on the day; API CPIC levels were used).
- Genotypes: chip file (GRCh38), Beagle 5.5 imputation with a public reference panel of complete genomes to 30x and verification of regions with bcftools.